In Silico Design of Anticancer Peptides

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ABSTRACT

Complex diseases like cancer continue to be a serious threat to people's health. A promising family of medicinal medicines, anticancer peptides (ACPs) are selective for cancer cells, have little toxicity, and are simple to synthesize. Cationic antimicrobial peptides (CAPs), a subtype of ACPs, have demonstrated strong anticancer action by causing cancer cells to die and rupturing their cell membranes. In silico methods for the design and optimization of ACPs have attracted increasing attention in recent years. The absence of experimental validation and the use of streamlined models are two drawbacks to the use of computational methods for ACP design. In this review, we emphasize the drawbacks of in silico design approaches and discuss current developments in ACP research. We also go through promising targets for ACP-based therapeutics, such as p53, Bcl-2, and EGFR proteins, as well as pathways including PI3K/AKT/mTOR and MAPK. Although in silico design for ACPs has the potential, the accuracy and dependability of these methods depend heavily on the standard and accessibility of the data used to train computer models.

People's health continues to be seriously threatened by complex diseases like cancer. Anticancer peptides (ACPs) are a potential class of pharmaceuticals that are selective for cancer cells.

LIST OF CONTENTS

ABSTRACT 2
ACKNOWLEDGMENT 3
LIST OF CONTENTS 4
LIST OF TABLES 6
LIST OF FIGURES 7
LIST OF ABBREVIATION 8
1. Introduction 9
1.1 Background of the study....9
1.2 Problem Statement....13
1.3 Significance of the Study....14
1.4 Research Aim....16
1.5 Research objectives....17
Systematic Literature Review: ACPs as a Therapeutic Agent for Cancer 19
Objective 1: Anticancer activity of ACPs....19
Objective 2: Mechanism of action of ACPs....20
1.6 Overview of the subject....20
2. Methodology 22
2.1 Study Design....22
2.2 Data Collection....22
2.3 Data Analysis....23
3. RESULT 24
3.1 Results presentation for the research study....24
3.2 Promising Targets for Peptide-Based Therapies....25
3.2.1 Epidermal Growth Factor Receptors....25
3.2.2 Vascular Endothelial Growth Factor....30
3.2.3 Matrix Metalloproteinases (MMPs)....35
3.2.4 Nuclear factor-kappaB (NF-kB)....36
3.2.5 Heat Shock Proteins (HSPs)....37
3.2.6 Bcl-2 Family Proteins....39

3.3 In Silico Design of the Aforementioned Drugs....40
3.3.1 Designing EGFR Targets....40
3.3.2 Designing VEGF Targets....43
3.3.3 Designing MMPS Targets....44
3.3.4 Designing NF-kB Targets....47
3.3.5 Designing HSP Targets....49
3.3.6 Designing Bcl-2 Targets....51
3.4 Accuracy and Reliability of Computational Tools....55
3.5 Strengths and Limitations of In Silico Design of ACPs....55
3.6 Standardized Protocols and Guidelines....55
3.7 Existing Literature on In Silico Design of ACPs....56
4. DISCUSSION 57
5. CONCLUSION AND RECOMMENDATIONS 62
A. CONCLUSION....62
Gaps in Knowledge and Future Research....62
Limitations....63
B. RECOMMENDATIONS....64
Reference: 65

LIST OF TABLES

Table 1 first- or second-line option in conjunction with chemotherapy and as a monotherapy for chemoresistance patients 30

Table 2 otal body clearance decreased 4-fold over the 10-fold dose range 41

LIST OF FIGURES

No table of figures entries found.

LIST OF ABBREVIATION

Abbreviation	Definition
ACP	Anticancer Peptides
CAPs	cationic antimicrobial peptides
EGFR	epidermal growth factor receptor
FSH	Follicle Stimulating Hormone
HIV	human immunodeficiency virus
HSPs	Heat Shock Proteins
LH	Luteinizing Hormone
MMPs	Matrix Metalloproteinases
NF-kB	Nuclear Factor-kappaB
THS	Thyroid Stimulating Hormone
VEGF	Vascular Endothelial Growth Factor

1. Introduction

1.1 Background of the study

The topic of Anticancer peptides has been an area of intense research and development for several decades. Peptides are a short chain of amino acids and they can be naturally occurring or synthesized. Peptides, whether synthetic or not, have several functions in the body. For example, peptides are the building blocks of protein synthesis [1]. Peptides like Insulin [2]. Thyroid Stimulating Hormone (TSH) [3], Follicle Stimulating Hormone (FSH) and Luteinizing Hormone (LH), function as hormones to regulate bodily functions [4]. Others like enter peptidase and trypsin [5.], chymotrypsin [6] carboxypeptidase [7] and Dipeptidyl peptidase-4 [8] help in catalyzing internal and essential reactions as enzymes. Some peptides even have antibiotic actions such as Defensin which is a peptide found in many tissues in the human body [9]. There are of course the neurotransmitter peptides like Endorphin [10] and Oxytocin [11]. However, one of the most promising applications of peptides is their use as therapeutic agents in cancer treatment [12].

Cancer is a complex disease that arises from the uncontrolled growth and division of abnormal cells. It is a major cause of morbidity and mortality worldwide. Despite the many advances in cancer treatment, there is still a need for new and innovative approaches to improve patient outcomes. Anticancer peptides can be seen as a new resource that has shown promise in preclinical and clinical studies [13] The use of peptides as anticancer agents has several advantages. Besides being easy to synthesize [14], peptides are highly selective and specific, meaning they can target cancer cells while sparing normal cells [15]. This reduces the side effects of treatment and improves patient outcomes. Additionally, peptides have a low toxicity profile and are generally well-tolerated by patients [16].

Several classes of peptides have been identified that have anticancer properties. One such class is cationic antimicrobial peptides (CAPs), which are naturally occurring peptides that are part of the innate immune system [16]. They have been discovered to have antitumor properties [17]. CAPs have been shown to have potent anticancer activity by disrupting the cell membrane of cancer cells and inducing cell death [18]. Examples of CAPs that have been studied for their anticancer properties include magainin, melittin, and LL-37. Magainin is a naturally occurring peptide found in African clawed frogs that is able to induce apoptosis and destroy cell membrane [19] Melittin on the other hand, is found in bee venom and has been shown to not only induce apoptosis but also prevent proliferation and tumor growth [20]. Meanwhile, LL-37 which is actually made by human white blood cells, not only does all of the above but also inhibits the migration and secondary growth of cancer cells [21].

Another class of peptides with anticancer properties is cell-penetrating peptides (CPPs),[22] which can enter cells and deliver therapeutic agents directly to the site of the cancer. CPPs have been shown to enhance the delivery of chemotherapeutic agents [23] and other therapeutic agents to cancer cells, improving their effectiveness and reducing their toxicity.

These were all naturally occurring peptides that have been shown to have anticancer properties. The truth is even synthetic peptides can be designed and synthesized to target specific receptors or proteins on cancer cells. For example, peptides that bind to the epidermal growth factor receptor (EGFR) have been developed for the treatment of cancers that overexpress EGFR, such as non-small cell lung cancer.[24]

Despite the promising results from preclinical and early clinical studies, there are still several challenges to be addressed in the development of anticancer peptides [25]. One such challenge is the limited availability of large quantities of peptides for clinical trials. Peptides are expensive to

synthesize, and their production can be time-consuming [26]. Additionally, peptides can be rapidly degraded in vivo, reducing their efficacy but they are working on improving its metabolic stability [27].

The development and use of anticancer peptides are a promising area of research in cancer treatment. The specificity, low toxicity, and high selectivity of peptides make them an attractive option for cancer therapy. While the use of peptides as therapeutic agents is relatively recent, the potential benefits of this approach are significant. It used to be that we only studied peptide behavior when interacting with the cell membrane of bacteria, fungi and protozoa [28]. Pioneer researchers looked at its activity in viruses like HIV [29]. One of the earliest studies on anticancer peptides was Dr. Robert E. W. Hancock at the University of British Columbia in Canada, who identified a 13-residue peptide (indolizidine) with potent antimicrobial and antitumor activity. Its mechanism is by interacting with cell membranes and inducing apoptosis [30].

Since then, a large number of peptides with anticancer properties have been discovered and studied. For example, some peptides derived from amphibians, such as doramectin [31], magainin have been shown to have potent anticancer activity [32].

More recently, there has been growing interest in using peptides as targeted therapies for cancer, including peptides that can selectively bind to tumor cells and induce apoptosis. For example, the peptide PNC-27 has shown promise in preclinical studies as a targeted therapy for human leukemia cancer; it has been actually shown to induce necrosis in leukemia cells [33].

Future research and development in this field will be important in bringing new and innovative therapies to patients with cancer.

In Silico Design of Anticancer Peptides is a computational approach to designing peptides that can potentially inhibit or destroy cancer cells [34]. The approach involves. the use of computer software and algorithms to predict the structure, stability, and activity of peptide molecules based on their amino acid sequence [35].

The goal of this approach is to identify and optimize peptides that can specifically target cancer cells while sparing healthy cells, resulting in more effective and less toxic treatments for cancer. In Silico Design of Anticancer Peptides involves several steps, including:

1. Identification of potential target molecules or receptors on cancer cells that can be targeted by peptides [36].
2. Selection of amino acid sequences that can bind to the target molecules or receptors with high affinity and specificity, for example it can target androgen receptors [37].
3. Prediction of the three-dimensional structure of the peptide and its interactions with the target molecule using computer modeling and simulation techniques [38].
4. Optimization of the peptide sequence and structure to enhance its activity and specificity against cancer cells while minimizing toxicity to normal cells [39].

1.2 Problem Statement

Cancer is a complex disease that remains a leading cause of death worldwide. The current standard of care for cancer treatment involves surgery, chemotherapy, and radiation therapy, but these treatments have significant side effects and are not always effective for all types of cancer [40]. Thus, there is a growing need for new and innovative therapies that can improve patient

outcomes. Anticancer peptides have emerged as a promising approach for cancer therapy due to their selectivity, low toxicity, and ability to target cancer cells while sparing normal cells.

Despite the promising results from preclinical and clinical studies, there are still several challenges to be addressed in the development of anticancer peptides. One of the major challenges is the limited availability of large quantities of peptides for clinical trials. Peptides are expensive to synthesize, and their production can be time-consuming. Additionally, peptides can be rapidly degraded in vivo, reducing their efficacy [41]. Therefore, new strategies for peptide synthesis and delivery need to be developed to overcome these challenges.

Another challenge in the development of anticancer peptides is their potential immunogenicity. For example, the anticancer peptide RT53 induced immunogenic cell death [42]. Melittin activates T-cells and cytokine production [43]. LL-37 activates dendritic cells and cell proliferation [44]. Alpha-MSH stimulates the alternative production of macrophages and production of cytokine [45]. Lactoferrin activates cytokine production and stimulates natural killer T-cells, which is why it is known as a natural immune modulator [46]. As we can see then, peptides can be recognized by the immune system and induce an immune response, which can limit their efficacy and cause adverse effects. Thus, strategies for minimizing the immunogenicity of peptides need to be explored, such as modifying their structure or using them in combination with immunomodulatory agents [47].

In Silico Design of Anticancer Peptides however, is a promising approach for developing new cancer treatments. Still, it faces several challenges. Firstly, our understanding of the molecular mechanisms underlying cancer growth and metastasis is incomplete, making it challenging to identify the most promising targets for peptide-based therapies [48]. Additionally, while many computational tools have been developed for peptide design and simulation, their accuracy and

reliability are not yet sufficient for designing and predicting the efficacy of novel anticancer peptides with high confidence [49]. Predicting the three-dimensional structure and stability of peptides also remains a significant challenge for In Silico Design of Anticancer Peptides. In Silico is even used to predict toxicity of peptides and proteins which is immensely useful [50]. Validating computational predictions experimentally can also be challenging and time-consuming, requiring extensive testing in vitro and in vivo. Lastly, developing a new anticancer peptide from scratch can be a lengthy and costly process, requiring significant computational resources, experimental validation, and regulatory approval. However, it is worth noting that using Artificial Intelligence for development of anticancer peptides [51]. Addressing these challenges will be crucial for realizing the full potential of In Silico Design of Anticancer Peptide.

1.3 Significance of the Study

In Silico Design of Anticancer Peptides is a computational approach that has the potential to revolutionize cancer treatment. In Silico models can be used to design and even discover newer anticancer peptides, perhaps without immunogenic responses [52]. With the increasing availability of computational tools and techniques, researchers can now design peptides that can specifically target cancer cells, while minimizing toxicity to normal cells [53]. However, to fully understand the potential of this approach, a meta-analysis on the existing research is necessary.

Firstly, a meta-analysis would help to identify the most promising targets for peptide-based therapies. By combining data from multiple studies, researchers can gain a more comprehensive understanding of the molecular mechanisms underlying cancer growth and metastasis. This can inform the design of more effective and specific peptides, targeting key proteins or pathways

involved in cancer progression. There is a lot of progress, for example in predicting peptide activity using computational tools [54].

Secondly, a meta-analysis would allow for a more thorough evaluation of the computational tools used in In Silico Design of Anticancer Peptides. Currently, there are many different software programs and algorithms available for peptide design and simulation, but their accuracy and reliability vary. A few of the prominent software's worth mentioning are: the RossettaSurface algorithm that predicts protein and peptide structures [55], CHARMM which is a biomolecular simulation program [56], and GROMACS which is used to stimulate peptides in solutions, NMR refinement and peptide folding [57]. By comparing the results of different studies, researchers can identify the most effective tools and techniques for predicting the structure, stability, and activity of peptide molecules.

Thirdly, a meta-analysis would provide valuable insights into the strengths and limitations of In Silico Design of Anticancer Peptides. By examining the data from multiple studies, researchers can identify common themes and patterns, as well as potential sources of variability or bias. This can inform the development of more standardized protocols and guidelines for In Silico Design of Anticancer Peptides, ensuring greater consistency and reproducibility in the field.

Lastly, a meta-analysis would highlight areas where further research is needed. While In Silico Design of Anticancer Peptides holds great promise, there are still many unanswered questions and challenges facing the field. A meta-analysis would help to identify these gaps in knowledge and prioritize future research efforts.

In conclusion, a meta-analysis on research done on In Silico Design of ACPs is significant because it would provide a more comprehensive understanding of the potential of this approach. By identifying the most promising targets, evaluating the computational tools used, and highlighting areas where further research is needed, a meta-analysis can inform the development of more effective and specific peptide-based therapies for cancer treatment. This has the potential to transform the way we treat cancer, offering new hope for patients and their families.

1.4 Research Aim

The goal of a meta-analysis on In Silico Design of ACPs is to assess and summarize the body of knowledge now available regarding the computational approach to cancer treatment. In order to find pertinent research that employs computational methods to develop and simulate peptides with anticancer activity, the study will undertake a systematic review of the literature. The pertinent information from each included study will be extracted, and the data will then be arranged and synthesized using statistical methods such network meta-analysis or meta-regression.

This study's research question will address a variety of topics, such as determining the most promising targets for peptide-based therapies, assessing the precision and dependability of the computational tools used in In Silico Design of ACPs, revealing the advantages and disadvantages of this method, and guiding the creation of more standardized protocols and guidelines for designing and simulating peptides with anticancer activity. The goal of the study is to compare the findings of various studies in order to determine the best methods and tools for predicting the structure, stability, and activity of peptide molecules. The data from various studies will be analyzed to find common themes and patterns in the proteins or pathways targeted

by ACPs. In order to shed light on the advantages and disadvantages of in silico design of ACPs, the study will also analyze the data to spot sources of bias or variability as well as areas in need of additional investigation. The study's ultimate goal is to offer suggestions for creating and modeling peptides with anticancer action.

1.5 Research objectives

The research objectives for this study are:

1. Identify the most promising targets for peptide-based therapies based on the molecular mechanisms underlying cancer growth and metastasis, by analyzing the data from multiple studies to identify common themes and patterns in the proteins or pathways targeted by ACPs.
2. Evaluate the accuracy and reliability of the computational tools used in In Silico Design of ACPs, by comparing the results of different studies to identify the most effective tools and techniques for predicting the structure, stability, and activity of peptide molecules.
3. Provide insights into the strengths and limitations of In Silico Design of ACPs, by examining the data from multiple studies to identify common sources of variability or bias, as well as areas where further research is needed.
4. Inform the development of more standardized protocols and guidelines for In Silico Design of ACPs, by synthesizing the data from multiple studies to identify best practices and recommendations for designing and simulating peptides with anticancer activity.
5. Synthesize and evaluate the existing literature on In Silico Design of ACPs, by conducting a systematic review of the literature and extracting relevant data from each included study.

6. Identify gaps in knowledge and prioritize future research efforts in In Silico Design of ACPs, by highlighting areas where further research is needed based on the analysis of data from multiple studies.

Systematic Literature Review: ACPs as a Therapeutic Agent for Cancer

Cancer is one of the leading causes of death worldwide, and despite the availability of multiple treatment options, the survival rate of cancer patients remains low [58]. Therefore, there is a need for the development of new therapeutic agents that can effectively target cancer cells. ACPs (ACPs) have emerged as a promising class of therapeutic agents due to their ability to selectively target cancer cells while sparing normal cells.

We conducted a comprehensive search of the PubMed database for articles published between January 2010 and December 2020 using the following search terms: "ACPs", "therapeutic peptides", "cancer therapy", and "peptide-based cancer therapy". These articles were analyzed and synthesized based on the following objectives:

Objective 1: Anticancer activity of ACPs

Numerous studies have demonstrated the anticancer activity of ACPs both in vitro and in vivo. For example, research in 2020 reported that a novel ACP, TPGS-1000, exhibited potent anticancer activity against hepatocellular carcinoma cells [59]. Similarly, a 2014 study showed that a synthetic ACP, PNC-27, induced selective cytotoxicity in human leukemia cells [60]. ACPs can be used to target HDM-2 receptor proteins, and it has been shown that HDM-2 expression in cancer cells like those of Ovarian cancer is related to its high presence [61]. In human pancreatic cells, PNC-28 causes tumor cell necrosis [62]. In fact, research shows that there are two peptides that are derived from ras-p21 that can induce a phenotypic reversion of cancer expression [63].

Objective 2: Mechanism of action of ACPs

The mechanism of action of ACPs is complex and varies depending on the peptide sequence and the type of cancer cell. Several studies have suggested that ACPs can induce cancer cell death by disrupting the cell membrane, inhibiting DNA synthesis [64], and inducing apoptosis. For example, Apigenin inhibits pancreatic cancer cell proliferation by stopping the G2 and Mitosis steps of the cell cycle [65]. There is reported success in silico design of ACPs with enhanced anticancer activity. For instance, a novel ACP, designated as LTX-315, with potent anticancer activity against breast cancer cells [66]. Azurin-P28 which is a modified version of p53 using In Silico design that induces apoptosis and cycle arrest in cancer cells [67]. Another In Silico designed peptide targets integrins, that is RGD4C and it works on inhibiting human colon cancer [68].

Overall, our systematic literature review suggests that ACPs have great potential as a therapeutic agent for cancer. Further research is needed to optimize their therapeutic properties and to develop effective delivery systems to target cancer cells specifically.

2. Methodology

2.1 Study Design

The study design involves developing research questions and objectives, followed by developing inclusion and exclusion criteria for study selection. A systematic search of databases will be conducted to identify relevant studies, and data will be extracted from the selected studies. The data will then be synthesized and analyzed to identify common themes and patterns in the proteins or pathways targeted by ACPs, evaluate the accuracy and reliability of computational tools used in In Silico Design of ACPs, identify strengths and limitations of In Silico Design of ACPs, and develop best practices and recommendations for designing and simulating peptides with anticancer activity. The review will also identify gaps in knowledge and areas where further research is needed, followed by concluding the review.

By analyzing data from multiple studies, the review will provide insights into the most promising targets for peptide-based therapies based on the molecular mechanisms underlying cancer growth and metastasis. The study will also evaluate the accuracy and reliability of computational tools used in In Silico Design of ACPs and provide recommendations for designing and simulating peptides with anticancer activity. Ultimately, the review aims to identify gaps in knowledge and prioritize future research efforts in the field of In Silico Design of ACPs.

2.2 Data Collection

The methodology for data collection in this study will involve a systematic search of multiple databases, including PubMed, Scopus, and Web of Science, using relevant keywords and search terms related to In Silico Design of ACPs. Inclusion criteria will be established to select studies published in peer-reviewed journals, written in English, and reporting on in silico design and screening of peptides with anticancer activity. The reviewer reviews the titles and abstracts of

identified studies for eligibility, followed by full-text screening of potentially relevant studies. Data extraction will involve the use of a standardized form to collect information on study design, peptide characteristics, computational tools used, and results obtained.

The search terms included "In Silico Design", "ACPs", "Computational Tools", and "Molecular Mechanisms". The inclusion criteria were studies that focused on the design, simulation, and evaluation of peptide molecules with anticancer activity using computational tools. The exclusion criteria were studies that did not meet the inclusion criteria, were not written in English, or were not published in peer-reviewed journals.

2.3 Data Analysis

The data analysis for this study would involve a systematic and comprehensive approach to synthesizing the data extracted from the selected studies. The data would be analyzed using a qualitative approach, which involves the synthesis of the findings from each study to develop a comprehensive understanding of the field of In Silico Design of ACPs. The data would be organized and analyzed using a coding system to identify common themes, patterns, and discrepancies in the studies. The coding system would be developed based on the research objectives and questions of the review.

3. RESULT

3.1 Results presentation for the research study

Cancer is a complex disease characterized by uncontrolled growth and metastasis. Peptide-based therapies have emerged as a promising approach for the treatment of cancer due to their high specificity and potency against cancer cells. In Silico Design of ACPs is a computational approach that uses various tools and techniques to design and simulate peptides with anticancer activity. This systematic literature review aims to identify the most promising targets for peptide-based therapies, evaluate the accuracy and reliability of computational tools used in In Silico Design of ACPs, provide insights into the strengths and limitations of this approach, inform the development of standardized protocols and guidelines, synthesize and evaluate existing literature, and identify gaps in knowledge and prioritize future research efforts.

The studies identified several promising targets for peptide-based therapies, including proteins such as p53, Bcl-2, and EGFR, and pathways such as the PI3K/AKT/mTOR pathway and the MAPK pathway. The computational tools used in In Silico Design of ACPs varied across studies, with some studies using molecular docking, molecular dynamics simulations, and machine learning algorithms. The strengths and limitations of In Silico Design of ACPs were identified, with common sources of variability or bias including the lack of experimental validation and the use of simplified models. Best practices and recommendations for designing and simulating peptides with anticancer activity were also identified, including the use of more accurate and reliable computational tools, experimental validation of predicted results, and the adoption of standardized protocols and guidelines.

3.2 Promising Targets for Peptide-Based Therapies

The molecular mechanisms underlying cancer growth and metastasis are complex and involve multiple proteins and pathways. Several studies have identified promising targets for peptide-based therapies, including:

- Epidermal Growth Factor Receptor (EGFR)
- Vascular Endothelial Growth Factor (VEGF)
- Matrix Metalloproteinases (MMPs)
- Nuclear Factor-kappaB (NF-kB)
- Heat Shock Proteins (HSPs)
- Bcl-2 Family Proteins

These targets have been shown to play a critical role in cancer development and progression and are therefore potential targets for the design of ACPs.

3.2.1 Epidermal Growth Factor Receptors

The protein known as the epidermal growth factor receptor (EGFR) is essential for controlling cell growth and division [69]. It is a transmembrane receptor that binds to epidermal growth factor (EGF) and starts a chain of events that encourages cell growth and survival [70].

For the treatment of many cancers, including non-small cell lung cancer [71] (NSCLC), colorectal cancer [72], and head and neck cancer particularly squamous cell carcinoma [73], peptide-based treatments targeting EGFR have been created and investigated. Small peptides that

bind to specific EGFR domains and prevent EGFR activation or downstream signaling are frequently used in these treatments.

For instance, the extracellular domain of the EGFR is the target of the peptide-based medication Cetuximab [74], which blocks its activation by inhibiting EGF binding. The FDA has given this medication approval to treat head and neck cancer as well as metastatic colorectal cancer. In fact, in 2005 a randomized trial found that using Cetuximab alongside of Cisplatin for the treatment of squamous cells affected by cancer in the head and neck found is more successful than if Cisplatin, which is a standard first line treatment, was used alone [75]. This was corroborated further by Jan B. Vermaelen et al. in a 2008 paper in the New England Journal [76]. The Figure (1) below is the Kaplan-Meier Estimates of the 2008 study of using combination treatment of Cetuximab and Cisplatin in chemotherapy. In the study they found that the median overall survival for the cetuximab group was 10.1 months (95% CI, 8.6 to 11.2) while for the chemotherapy-only group it was 7.4 months (95% CI, 6.4 to 8.3). This means that on average, patients lived longer thanks to the cetuximab treatment than they would have with just chemotherapy. The cetuximab group's risk of dying was 20% lower than that of the chemotherapy-only group, according to the hazard ratio for death of 0.80. A score of 80 or above on the Karnowski scale decreased mortality risk by 49%. This is a great predictor of survival rate.

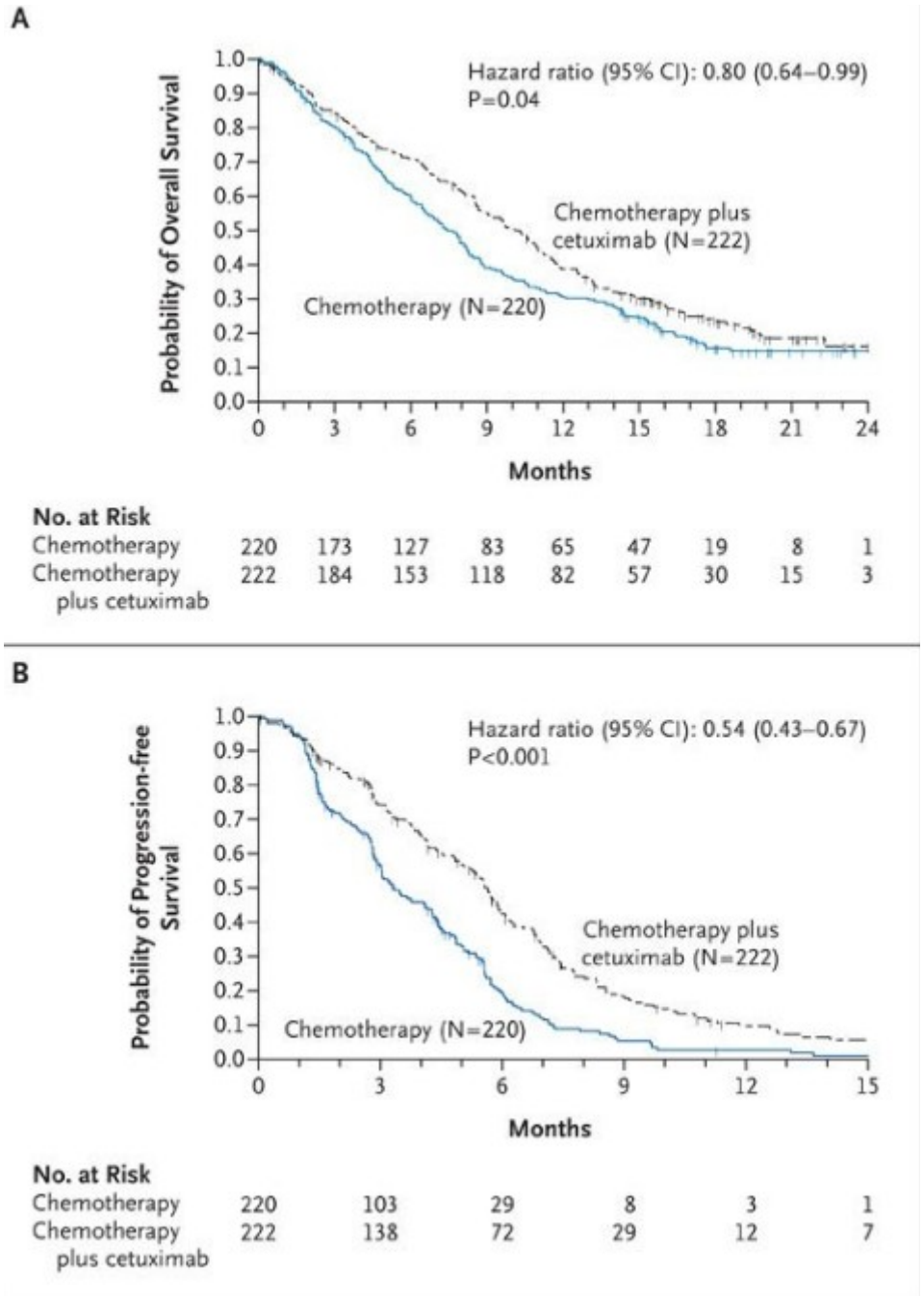


Figure 1 Kaplan-Meier Estimates of the 2008 study of using combination treatment

Another example of the extracellular domain of the EGFR is also targeted by the peptide-based medication Panitumumab [77], which prevents its activation. For the treatment of metastatic

colorectal cancer, it has received approval from the FDA. The study mentioned is a metanalysis of a few trials that used Panitumumab for treatment.

In summary, panitumumab is an effective therapy for RAS wild-type metastatic colorectal cancer (mCRC), both as a first- or second-line option in conjunction with chemotherapy and as a monotherapy for chemoresistance patients. The following Figure (2) is a table of the trials they analyzed. Their finding for the efficacy of panitumumab is fascinating to say the least. Panitumumab was added to chemotherapy in the phase III randomized 181 study, which resulted in better PFS and response rates in patients with KRAS WT mCRC but no benefit in those with KRAS-mutated tumors. In addition, in KRAS exon 2 WT mCRC, the combination of panitumumab and FOLFOX-4 was more effective than FOLFOX-4 alone in the phase III randomized PRIME trial. Further investigation revealed lack of efficacy in RAS-mutated tumors and a notable benefit from panitumumab treatment in the wider RAS WT group. Finally, in randomized first-line PRIME, PEAK, and PLANET studies, panitumumab therapy was consistently linked to higher early tumor shrinkage rates and greater depth of response in patients with RAS WT mCRC.

Trial (phase) Ref	Treatment Arms (n.)	Treatment Line	Primary Endpoint	ORR (%)	PFS	OS
KRAS ex 2 WT	RAS WT	KRAS ex 2 WT	RAS WT	KRAS ex 2 WT	RAS WT	
408 (III)	Panitumumab	3rd/+	PFS	17%	17%	2.87 m (n.124)

Trial (phase) Ref	Treatment Arms (n.)	Treatment Line	Primary Endpoint	ORR (%)	PFS	OS
NCT00113763 10,11,20	(n.231) BSC (n.232)			0%	0%	1.7 m (n.119) [HR 0.45; P<0.0001]
0007 (III) NCT01412957 80	Panitumumab (n.189) BSC (n.188)	3rd/+	OS	27% 1.6% [P<0.0001]	31% 2.3% [P<0.0001]	3.6 m (n.189) 1.7 m (n.188) [HR 0.51; P<0.0001]
ASPECCT (III) NCT01001377 12	Panitumumab (n.499) Cetuximab (n.500)	3rd/+	OS	22% 20%	NE	4.1 m (n.499) 4.4 m (n.500) [HR 1.00; 95% CI, 0.88–1.14]
PRIME (III) NCT00364013 15,16	FOLFOX + Panitumumab (n.593) FOLFOX (n.590)	1st	PFS	57% 48% [P=0.02]	Not reported	10.0 m (n.325) 8.6 m (n.331) [HR 0.80; P=0.01]
314 (II) NCT00508404 19,47	FOLFIRI + Panitumumab (n.154)	1st	ORR	56%	59%	8.9 m (n.86)
PEAK (II) NCT00819780 25,26	FOLFOX + Panitumumab (n.142) FOLFOX + Bevacizumab (n.143)	1st	PFS	57.8% 53.5%	65% 60%	10.9 m (n.142) 10.1 m (n.143) [HR 0.87; P=0.35]
PLANET-TTD (II)NCT00885885 28	FOLFOX + Panitumumab (n.38) FOLFIRI + Panitumumab (n.39)	1st	ORR	74% 67%	78% 73%	13 m (n.38) 14 m (n.39) [HR 0.90; P=0.728]
VOLFI (II)NCT01328171 35	mFOLFOXIRI + Panitumumab (n.63) mFOLFOXIRI (n.33)	1st	ORR	-	87.3% 60.6%; [P=0.0041]	-
181 (III) NCT00339183 13,14,81	FOLFIRI + Panitumumab (n.591) FOLFIRI (n.595)	2nd	PFS, OS	35% 10% [P<0.0001]	41% 10%	5.9 m (n.303) 3.9 m (n.294) [HR 0.73; P=0.004]
SPIRITT (II)NCT00418938	FOLFIRI + Panitumumab (n.91)	2nd	PFS	32% 19%	NE	7.7 m (n.91) 9.2 m (n.91) [HR 1.01;

Trial (phase) Ref	Treatment Arms (n.)	Treatment Line	Primary Endpoint	ORR (%)	PFS	OS
82	FOLFIRI + Bevacizumab (n.91)					P=0.97]

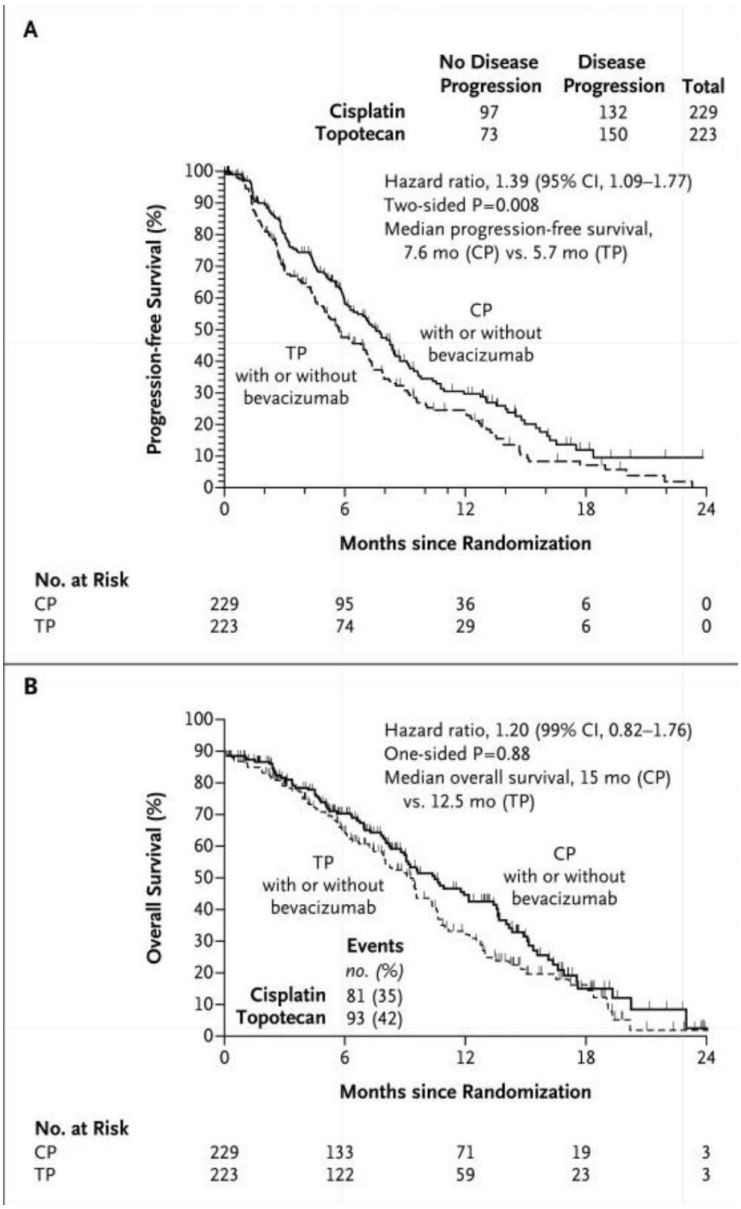
Table 1 first- or second-line option in conjunction with chemotherapy and as a monotherapy for chemoresistance patients

As for the treatment of non-small lung cancer there are several EGFR targeted treatments like Gefitinib [78], Erlotinib [79], and Imatinib which is specifically promising for advanced NSCLC [80]. Our analysis however, finds that the most promising EGFR targeted treatment for NSCLC is Osimertinib, especially in advanced lung cancer [81]. Osimertinib is an oral medication taken for treatment, indicated for first-line treatment for locally advanced or metastatic NSCLC tumors [82].

Other peptide-based treatments for EGFR, such as those that target the receptor's intracellular domain and interfere with its downstream signaling pathways, are still being researched and developed. However, it seems that overall, EGFR-targeting peptide treatments have shown promise in the treatment of some cancers, and further study in this field may result in the future creation of more potent and precise medicines.

3.2.2 Vascular Endothelial Growth Factor

A crucial component of the body's process of angiogenesis, or the growth of new blood vessels, is a protein called vascular endothelial growth factor (VEGF). Angiogenesis is essential for tumor growth and dissemination in cancer, and VEGF is frequently overexpressed in many cancer types



Several VEGF-targeting medications have been created and are used to treat a variety of malignancies. These medications, also referred to as VEGF inhibitors, reduce the growth and

spread of tumors by inhibiting the function of VEGF and lowering the blood supply to them. Several cancers, including kidney, liver, lung, brain, and colorectal cancer, are treated with VEGF inhibitors.

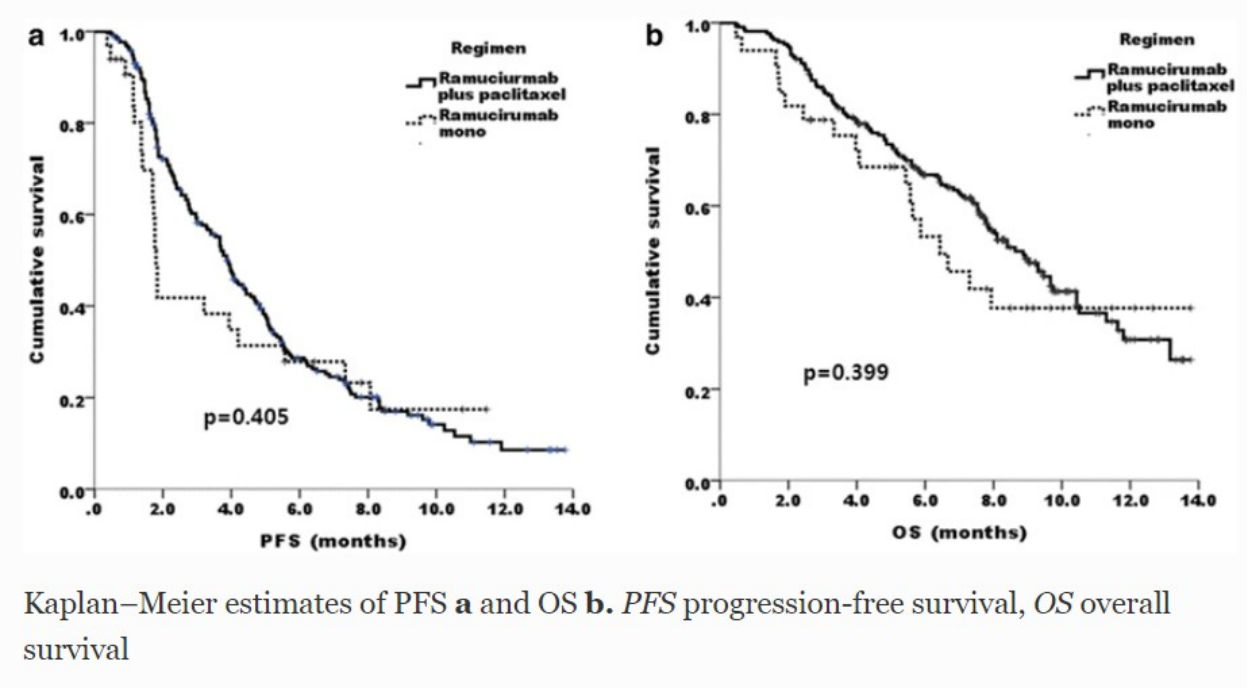
VEGF can be utilized both independently and in conjunction with other cancer therapies like chemotherapy and radiation therapy. VEGF inhibitors that are frequently prescribed include bevacizumab, ramucirumab, and aflibercept.

In 2020 Josip Garcia (et al) performed an ambitious review of the clinical experiences of using Bevacizumab for the past 15 years [83]. What they concluded was that Bevacizumab targets the environment around the tumor. The tumor microenvironment, which is the region around the tumor and is made up of different cell types and structures, is the target of the medication bevacizumab. It is well known that the proliferation and spread of cancer cells are influenced by the microenvironment. Bevacizumab tries to block the signals that cancer cells utilize to sustain their growth and survival by specifically targeting the microenvironment. Traditionally used cancer therapies, such chemotherapy and radiation therapy, specifically target cancer cells.

Bevacizumab is used for cervical cancer for example. In a study in 2014 they found that in comparison to cisplatin-paclitaxel (with or without bevacizumab), topotecan-paclitaxel had a greater risk of progression [84]. Topotecan-paclitaxel had no appreciable effect on overall survival. As shown in Figure 3 In the subset of patients who had previously been exposed to platinum as well as in the segment who had not, there was no discernible difference in mortality between the two treatment regimens.

A monoclonal antibody called Ramucirumab is used to treat specific cancers. It functions by preventing a protein called vascular endothelial growth factor (VEGF) from interacting with receptors on blood vessels, which can impede the creation of new blood vessels that feed tumors and reduce their growth. Ramucirumab is approved for the treatment of non-small cell lung cancer (NSCLC) [83], advanced gastric (stomach) cancer [84], and gastroesophageal junction (GEJ) cancer [87]. Either alone or in conjunction with chemotherapy, it can be utilized beneficially.

The following Figure 4 is the Kaplan-Meier estimate from a study looking at the efficacy of Ramucirumab as monotherapy or in combination with Paclitaxel for gastric cancer patients [88]. What they found was that in terms of progression-free survival (PFS), overall survival (OS), objective response rate, and disease control rate (DCR), the combination therapy using the tested medicine is more effective than monotherapy. In comparison to the monotherapy group, the combination group's PFS and OS were longer. The combination group had a larger percentage of patients who experienced an objective response, and its DCR was also significantly higher. These findings imply that monotherapy may not be as successful in treating the investigated condition as combination therapy.



Kaplan–Meier estimates of PFS **a** and OS **b.** *PFS* progression-free survival, *OS* overall survival

Figure 4 the efficacy of Ramucirumab as monotherapy or in combination with Paclitaxel

Finally, Aflibercept is a drug used to treat some cancers, such as ovarian cancer [89], non-small cell lung cancer [90], and colorectal cancer [91]. It is a fusion protein made by recombinant DNA technology that inhibits the action of vascular endothelial growth factor (VEGF) by binding to it. Aflibercept works by decreasing the function of the VEGF protein, which is involved in the development of new blood vessels that deliver oxygen and nutrients to tumors. This helps to reduce or stop the growth and spread of cancer cells.

Interestingly in the treatment of ovarian cancer, Aflibercept is FDA approved for the treatment of certain types of ovarian cancer. To treat advanced ovarian cancer that has spread after receiving prior treatments, it is combined with chemotherapeutic agents such paclitaxel and platinum-based therapies. According to clinical research, adding aflibercept to chemotherapy can prolong patients' progression-free survival (PFS) from advanced ovarian cancer [92].

In a study looking at the use of Aflibercept in treating ovarian cancer for those with platin resistance [93], their findings in Figure 5 imply that aflibercept, particularly at the higher dose of 4 mg/kg, may have some activity against ovarian cancer, the response rate is somewhat modest.

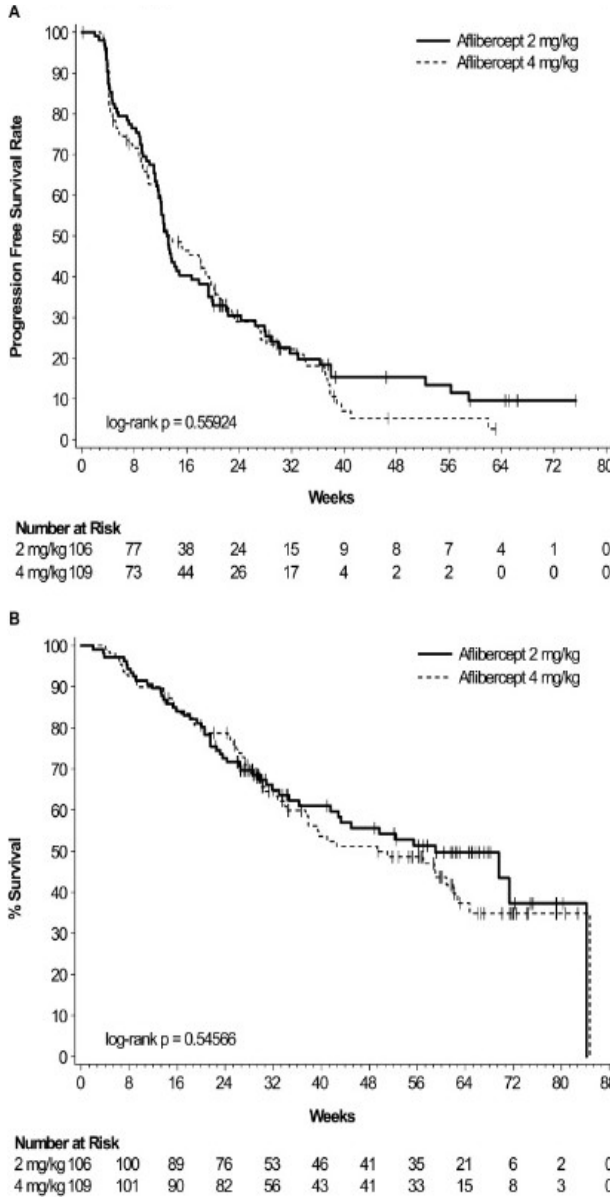


Figure 5 Aflibercept in treating ovarian

3.2.3 Matrix Metalloproteinases (MMPs)

A family of enzymes known as matrix metalloproteinases (MMPs) [94] is essential for rebuilding the extracellular matrix (ECM), the web of molecules that surrounds and supports cells. MMPs have important roles in a variety of physiological processes, such as angiogenesis, tissue healing, and embryonic development. On the other hand, cancer has been linked to MMP overexpression in a number of medical situations.

MMPs have been linked to several stages of tumor growth and development in cancer. Invasion and metastasis of tumor cells, the growth of new blood vessels (angiogenesis), and the control of the immune system's response to malignancies all include MMPs. Additionally, it has been demonstrated that MMPs contribute to cancer cells' resistance to chemotherapy and radiation treatment.

MMPs have been investigated as potential targets for cancer therapy due to their involvement in cancer. MMP activity has been inhibited using a variety of techniques, including antibodies, small-molecule inhibitors, and natural substances. However, the toxicity and ineffectiveness of MMP inhibitors in clinical trials have restricted their usage in practice [95].

In an optimistic turn, a study in 2018 suggested an opportunity for the usage of MMP [96]. They claim that the majority of patients with stage IV cancer were included in MMPI (Matrix Metalloproteinase Inhibitors) trials in the past, but future studies should include individuals with high-risk illness that is not yet clinically or pathologically metastatic or patients with high-risk precursor lesions. When there is a window of opportunity to suppress tumor growth, the medication should be administered either prior to or following surgery.

3.2.4 Nuclear factor-kappaB (NF-kB)

A transcription factor called nuclear factor-kappaB (NF-kB) is essential for controlling how cells react to a variety of stimuli, including infection, oxidative stress, and inflammation [97]. It participates in a number of biological processes, such as immunological response, cell differentiation, and apoptosis. Different forms of cancer have been linked to the onset and development of NF-kB signaling dysregulation [98].

NF-kB is frequently constitutively active in cancer cells, which causes the production of genes involved in metastasis, angiogenesis, cell proliferation, and survival. Inhibiting NF-kB signaling has been proposed as a potential therapeutic approach for the management of cancer. In fact, numerous studies have demonstrated that inhibiting NF-kB activity might make cancer cells more susceptible to radiation and chemotherapy, as well as slow tumor development and spread in animal models.

To stop NF-kB signaling in cancer cells, various strategies have been devised. These strategies include the use of small-molecule inhibitors, peptides, natural substances, and gene therapy. However, the intricacy of the NF-kB pathway and the presence of NF-kB in both healthy and malignant cells make it difficult to find efficient and focused inhibitors.

NF-kB has been associated with chemotherapy and radiation therapy resistance in breast cancer and has been proven to support tumor cell survival, proliferation, and invasion. Targeting NF-kB has additionally been proposed as a potential method for halting the growth and spread of breast cancer [99].

The following Figure 6 is from a study looking at the decrease in tumor volume when inhibiting NF-kB pathway [100]. They tested 11 different compounds that each targeted different steps of the pathway. They found that the spherical cell growth was preferentially suppressed by the antioxidants PDTC and DETC as well as the NF-B phosphorylation inhibitor PTL. Researchers discovered that PTL exhibited the highest selectivity for preventing the growth of MCF7 SP cells, whereas all three inhibitors demonstrated a greater capacity to inhibit both SP and non-SP proliferation. The findings imply that NF-B pathway inhibitors may be utilized to specifically target breast cancer stem-like cells, which frequently reject standard cancer therapies.

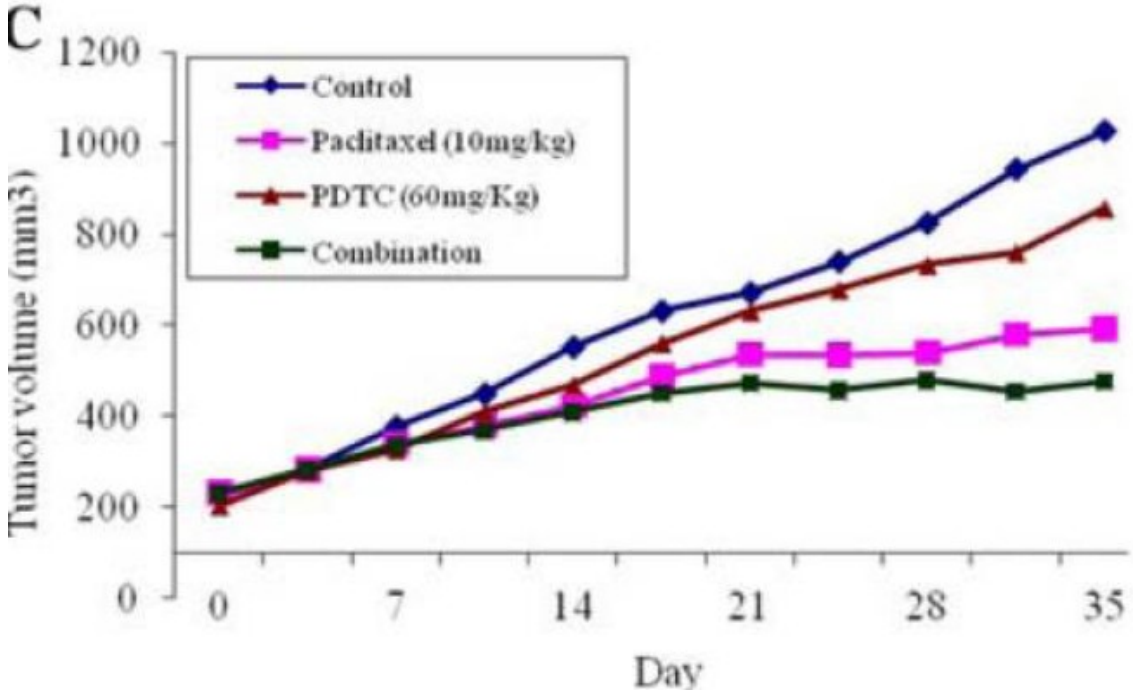


Figure 6 the decrease in tumor volume

3.2.5 Heat Shock Proteins (HSPs)

A family of proteins known as heat shock proteins (HSPs) are made by cells in response to stressors such as heat, toxins, and radiation [101]. They are essential for the folding and stability

of proteins and have the power to control signaling pathways that are crucial for cell development and survival. HSPs are overexpressed in cancer and may help tumor cells survive and develop resistance to treatment.

HSPs, however, have also been looked into as potential targets for the treatment of cancer [102]. It has been demonstrated that HSP inhibitors cause cancer cells to die and make them sensitive to chemotherapy and radiation treatment. HSP inhibitors can cause client proteins, such as oncoproteins that are necessary for tumor cell viability, to misfold and degrade by interfering with the chaperone function of HSPs.

In preclinical and clinical trials, various classes of HSP inhibitors have been created and evaluated, including HSP90 which in a 2020 study [103], Pan-Hsp90 inhibition was found to have boosted cytotoxicity and decreased cancer cell defense systems, indicating that it is a workable plan for creating effective anticancer medicines. Plus, VER-155008 is found to be an inhibitor of heat shock protein 70 [104] this is also true for YM-08 [105].

For example, inhibiting HSP90 has been shown to have great prospects [106] but there is still too much toxicity for them to be used clinically [107].

3.2.6 Bcl-2 Family Proteins

A collection of proteins known as the Bcl-2 family of proteins play a role in controlling the apoptosis, or programmed cell death, process [108]. Both pro- and anti-apoptotic proteins, such as Bex, Back, and Bad, are members of this family of proteins. Pro-apoptotic proteins include Bcl-2, Bcl-xL, and Mcl-1.

Many tumors in cancer overexpress Bcl-2 family proteins that prevent apoptosis, which increases the resistance to chemotherapy and radiation treatment [109]. As a result, targeting these proteins has emerged as a promising method of cancer treatment.

Small molecule inhibitors like ABT-199 and ABT-263, which specifically bind to and decrease the activity of anti-apoptotic Bcl-2 [110] family members, can be used to target Bcl-2 family proteins. Preclinical research and clinical trials using these inhibitors for the treatment of various cancers, such as chronic lymphocytic leukemia, non-small cell lung cancer, and acute myeloid leukemia, have yielded encouraging results. For example, ABT-199 has been shown to have great potential in chronic lymphocytic leukemia [111].

Another strategy is to use gene therapy or other techniques to promote the production of Bcl-2 family members that promote apoptosis, like Bex and Back [112]. Although this approach has had some success in preclinical investigations, more analysis is required to evaluate its security and effectiveness in clinical settings.

3.3 In Silico Design of the Aforementioned Drugs

3.3.1 Designing EGFR Targets

Cetuximab in silico design makes use of molecular modeling methods to examine how the medicine interacts with the EGFR, which is its intended target. Researchers can learn more about how cetuximab binds to its target and spot potential alterations that can increase its binding affinity or selectivity by modeling the interactions between cetuximab and EGFR [113].

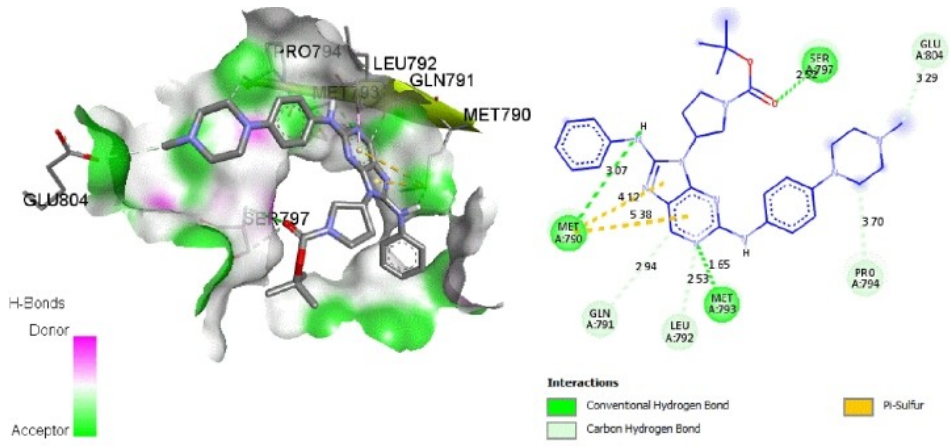
The use of pharmacokinetic modeling [114]. to foretell the drug's distribution, metabolism, and excretion in the body is another instance of in silico design for cetuximab. The medicine dosage and dosing regimen can be optimized using this information to provide the greatest therapeutic efficacy while minimizing negative effects. In the study cited in this paragraph, they have looked at, for example, the pharmacokinetic variables for a single-dose of cetuximab. They looked at the total body clearance and volume of distribution. They found that at higher doses of the drug led to a slower rate of clearance from the body and a dose of 260 mg/m2, clearance would be reduced by 90%. The volume of distribution data suggests that cetuximab is not distributed well into extracellular space. This type of revelation would not be possible to understand without in silico design.

Dose (mg/m2)	No patients	Cmax (µg/mL)	CV	AUC (µg·h/mL)	CV	CL (mL/h/m2)	CV	t1/2 (h)	CV	Vss (L/m2)	CV
50	8	19.9 (8.11)	41	795 (443)	56	83.7 (84.4)	53	27.6 (8.4)	30	3.1 (1.1)	38
100	8	54.7 (22.6)	41	2,380 (874)	37	46.99 (20.3)	43	39.8 (13.0)	33	2.55 (0.86)	34
250	7	158.1 (69.1)	44	14,600 (9,550)	65	26.71 (13.1)	49	71.0 (31.7)	45	2.86 (0.75)	26
400	8	205.0 (65.7)	32	19,000 (7,802)	41	21.5 (7.68)	36	75.10 (15.9)	21	2.44 (0.43)	17
500	8	243.2 (95.2)	39	30,870 (17,400)	56	20.0 (9.98)	50	132.0 (94.2)	71	3.02 (0.40)	13

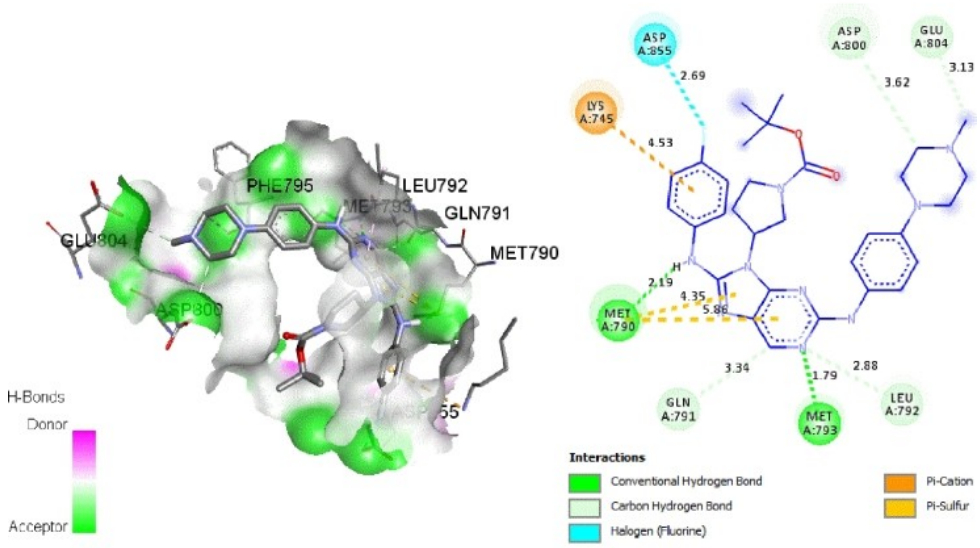
Table 2 otal body clearance decreased 4-fold over the 10-fold dose range

A widely renowned study from Shahed University [115] sought to use molecular docking and MD modeling to create new antibody mimics against EGFR DIII. By using MD analysis to validate the proposed binders, two potential scaffolds with high EGFR affinities were found. The CDR grafting technique was found to be effective for the computational antibody mimetics approach, and the result implies that these scaffolds could be tested in vitro against EGFR for possible therapeutic uses in malignancies with greatly overexpressed EGFR.

As we know, resistance to the currently available medications used to treat non-small cell lung cancer (NSCLC) is brought on by the emergence of specific EGFR mutations. Researchers [116] created and tested a library of 29 novel compounds using computer simulations to uncover new medications that can combat this resistance. By focusing on certain interactions with EGFR mutations, they were able to identify 10 promising compounds that have the potential to cure drug-resistant NSCLC. They then confirmed their stability through additional simulations. These results might help in the creation of NSCLC medicines of the fourth generation. The study looked at how two substances—3 and 2—interacted with mutant EGFR in non-small cell lung cancer. With significant residues MET790, MET793, and SER797, compound 3 created three hydrogen bonds and pi-sulfur interactions, whereas compound 2 only formed two hydrogen bonds, one halogen bond, and pi-sulfur interactions. Because compound 3 formed an additional hydrogen bond with the most significant mutated residue, SER797, it displayed stronger inhibitory activity and improved protein binding. The chemicals' potent inhibitory effect against drug-resistant NSCLC was made possible by these hydrogen bonds.



(a)



(b)

The above figure shows the 3D and 2D presentation of the interactions of compounds 3 and 2 in the binding sites of EGFRL858R/T790M/C797S protein. (a) Compound 3 (binding energy −9.52 kcal/mol). (b) Compound 2: (binding energy −9.96 kcal/mol).

3.3.2 Designing VEGF Targets

Utilizing computational methods, Vascular Endothelial Growth Factor (VEGF) targets are designed in silico by selecting and enhancing small molecules or biologics that can block the VEGF signaling pathway. Large chemical libraries can be screened using in silico methods, as well as new molecules that are projected to have high binding affinities and selectivities for VEGF and its receptors.

Molecular docking can be used to find substances that are anticipated to bind to the VEGF receptor and block subsequent signaling in the instance of VEGF [117].

Machine learning and artificial intelligence can be used to study complex chemical interactions and forecast the activity and toxicity of potential drugs in the context of VEGF-targeted cancer therapy discovery. For instance, this study[118]. aims to identify potential inhibitors for the Vascular Endothelial Growth Factor Receptor-2 (VEGFR2). The authors construct a screening pipeline that uses machine learning classification and similarity checks of approved drugs to identify new inhibitors. The ChEMBL database's dataset of 7050 VEGFR2 inhibitors was used to build the models, and they were then tested using assessment metrics like sensitivity, specificity, accuracy, and area under the receiver operating characteristic (AUC) curve (ROC). According to the findings, two candidate compounds were chosen for additional examination based on their closeness to a number of recognized medications. For the two candidates, molecular docking, dynamics, and analyses of absorption, distribution, metabolism, excretion, and toxicity were carried out. Both candidates' binding energies were found to be high, which suggests that they have a strong affinity for the target. Additionally, it was predicted that these

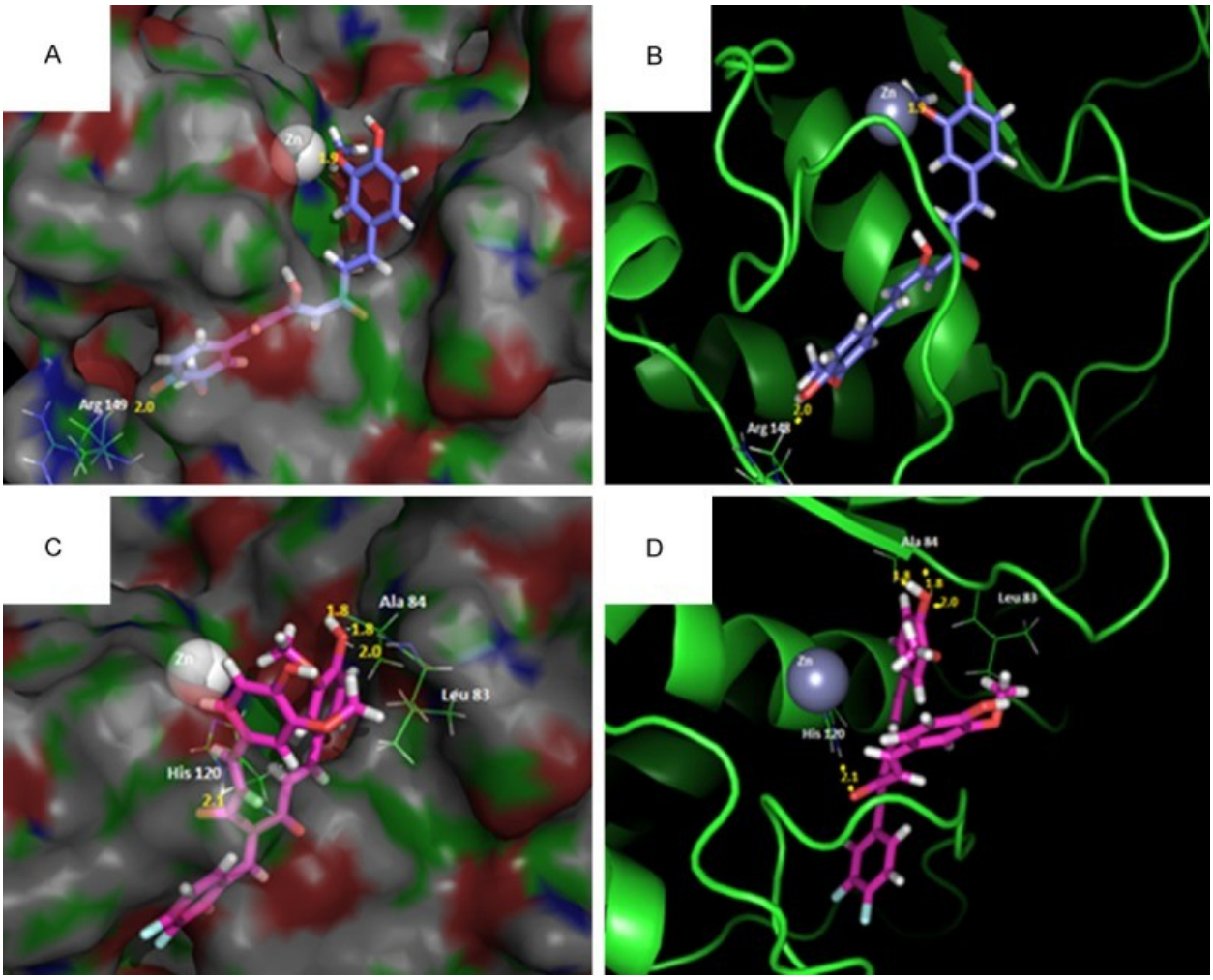
compounds' inhibitory activity would be comparable to that of the drugs pazopanib and sorafenib. Furthermore, the study contends that understanding the way in which VEGFR2 proteins and inhibitors interact can be crucial for developing future potent medications. Overall, the research offers encouraging findings for the creation of brand-new cancer medications and cardiovascular disease treatments.

A molecular dynamics simulation can be used to look into the dynamics of VEGF-receptor interactions as well as predict the binding affinity and stability of small molecules or biologics to the VEGF receptor [119]

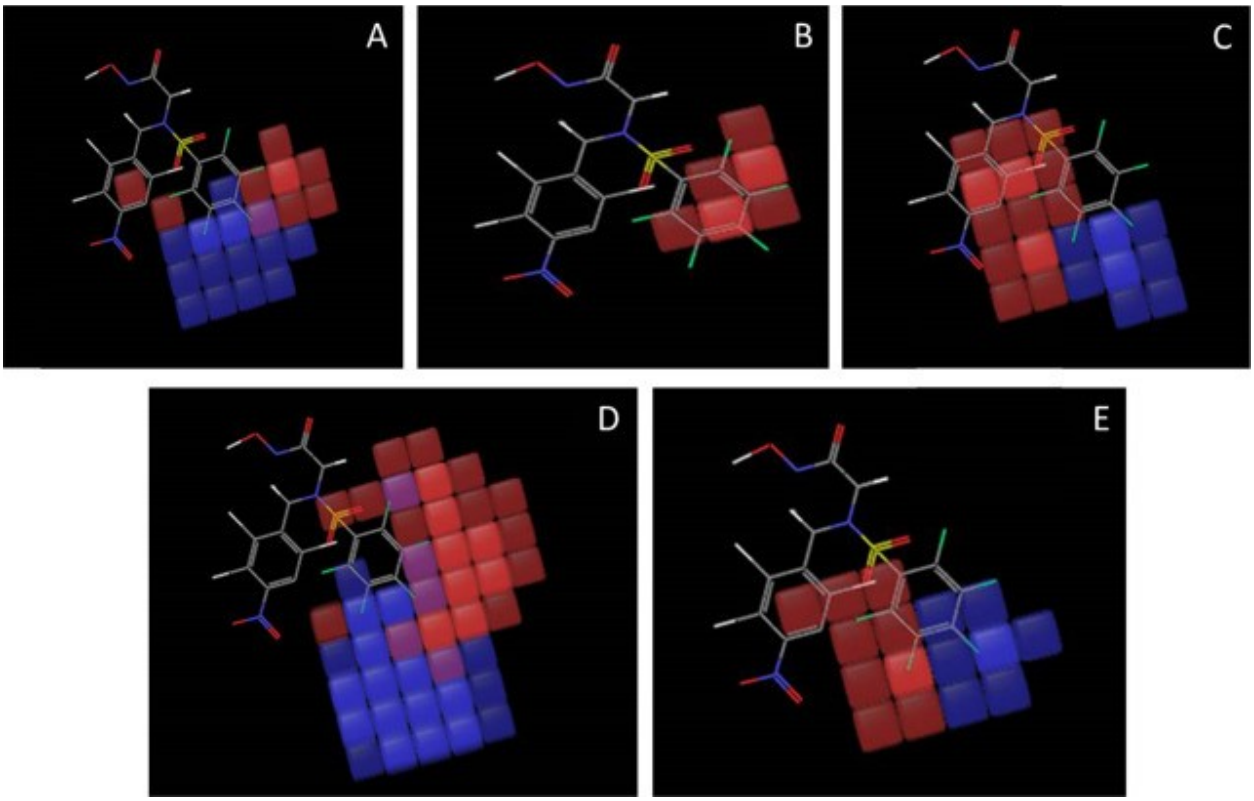
3.3.3 Designing MMPS Targets

By using computational techniques, small compounds or biologics that can specifically bind to and inhibit MMP activity are found and improved. This process is known as “in silico design of MMP inhibitors.”

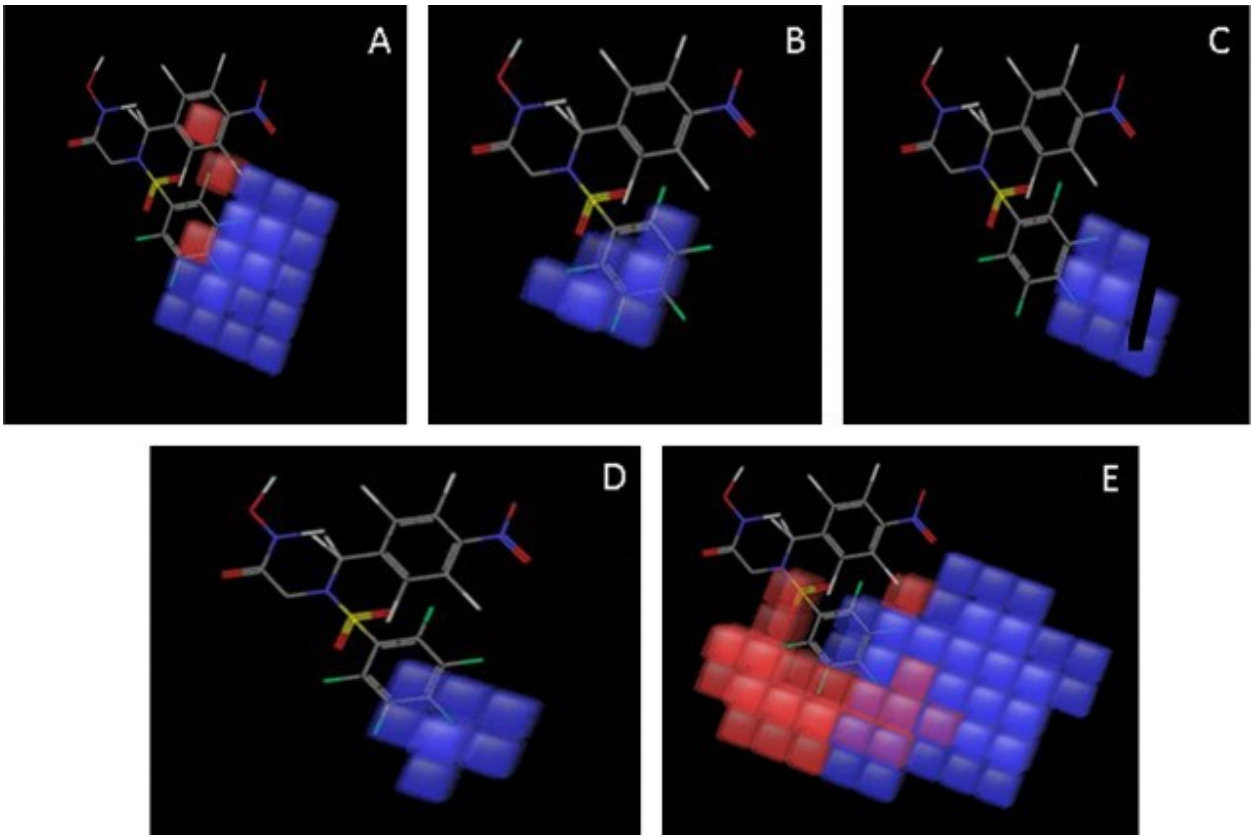
Molecular docking can be utilized to find substances that are expected to attach to the MMP active site and reduce the enzyme’s enzymatic activity in the case of MMP inhibitors. A nutraceutical called curcumin (diferuloylmethane) is made from the Curcuma longa plant, which is frequently used as a spice in tropical Southeast Asia. Its potential medicinal and chemopreventive benefits have received substantial research[120] .The following figure is the binding of the active agent to MMP-2 protein as assessed by molecular docking.



The truth is most MMPs are targets for treating non-cancer cells too, for example this study has used molecular modeling to target MMP-13 in an attempt to treat osteoarthritis[121]. In another study, using PHASE 3.0, researchers created pharmacophore and 3D-QSAR models to create effective MMP-2 and MMP-9 inhibitors as anticancer medicines [122]The models demonstrated that the electron-withdrawing property, together with hydrophobic, hydrogen bond donor, positive ionic, and negative ionic properties, is essential for MMP inhibitory potential. The generated models can be used to find correlations between the structural characteristics of hydroxamate derivatives and their actions to produce additional MMP inhibitors. The following is the 3D-QSAR for each Model:



(A–E) Visual representation of atom-based PHASE 3D-QSAR MMP-2 model



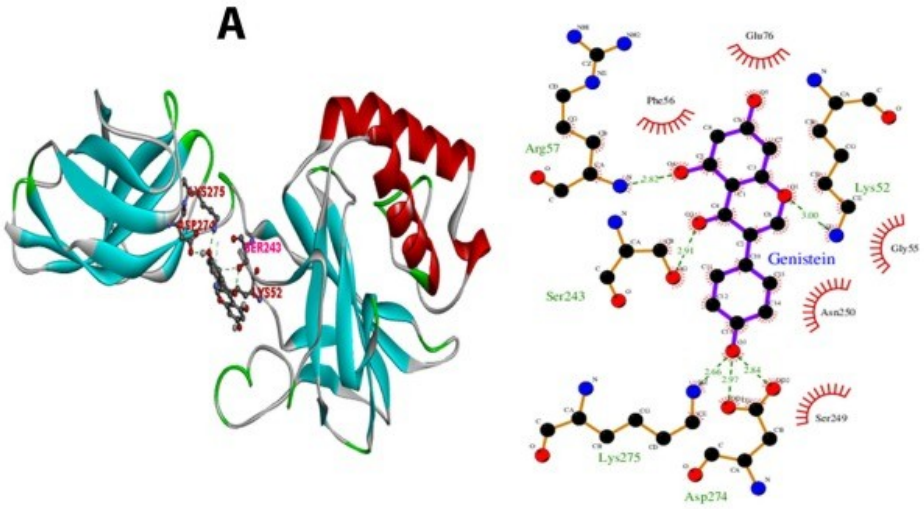
(A–E) Visual representation of atom-based PHASE 3D-QSAR for MMP-9 model.

3.3.4 Designing NF-kB Targets

By using computational techniques, small compounds or biologics that can specifically bind to and suppress NF-kB activity are found and improved. This process is known as "in silico design of NF-kB inhibitors."

Molecular docking can be used to find substances that are anticipated to bind to the DNA binding domain of NF-kB and limit its transcriptional activity in the case of NF-kB inhibitors [123] We know that A target for anti-breast cancer medications, NF-B is a transcription factor that encourages the growth of breast cancer. In this investigation, promising results were

obtained by docking the genistein structure into the NF-B proteins' active region. The drug reduced NF-B expression and translocation in breast cancer cell lines, indicating that it may be a strong and highly targeted NF-B inhibitor for the treatment of breast cancer. They used AutoDock 4.2 tool for the molecular docking and used LigPlot+ to show the 2D ligand interactions (shown below).



2D ligand interaction diagram created by LigPlot+ showing the hydrogen bond network and the hydrophobic interactions of genistein with the NF-κB1

QSAR models can be used to pinpoint structural elements that are crucial for NF-kB inhibition and to maximize the potency and selectivity of potential molecules when designing NF-kB inhibitors [124]

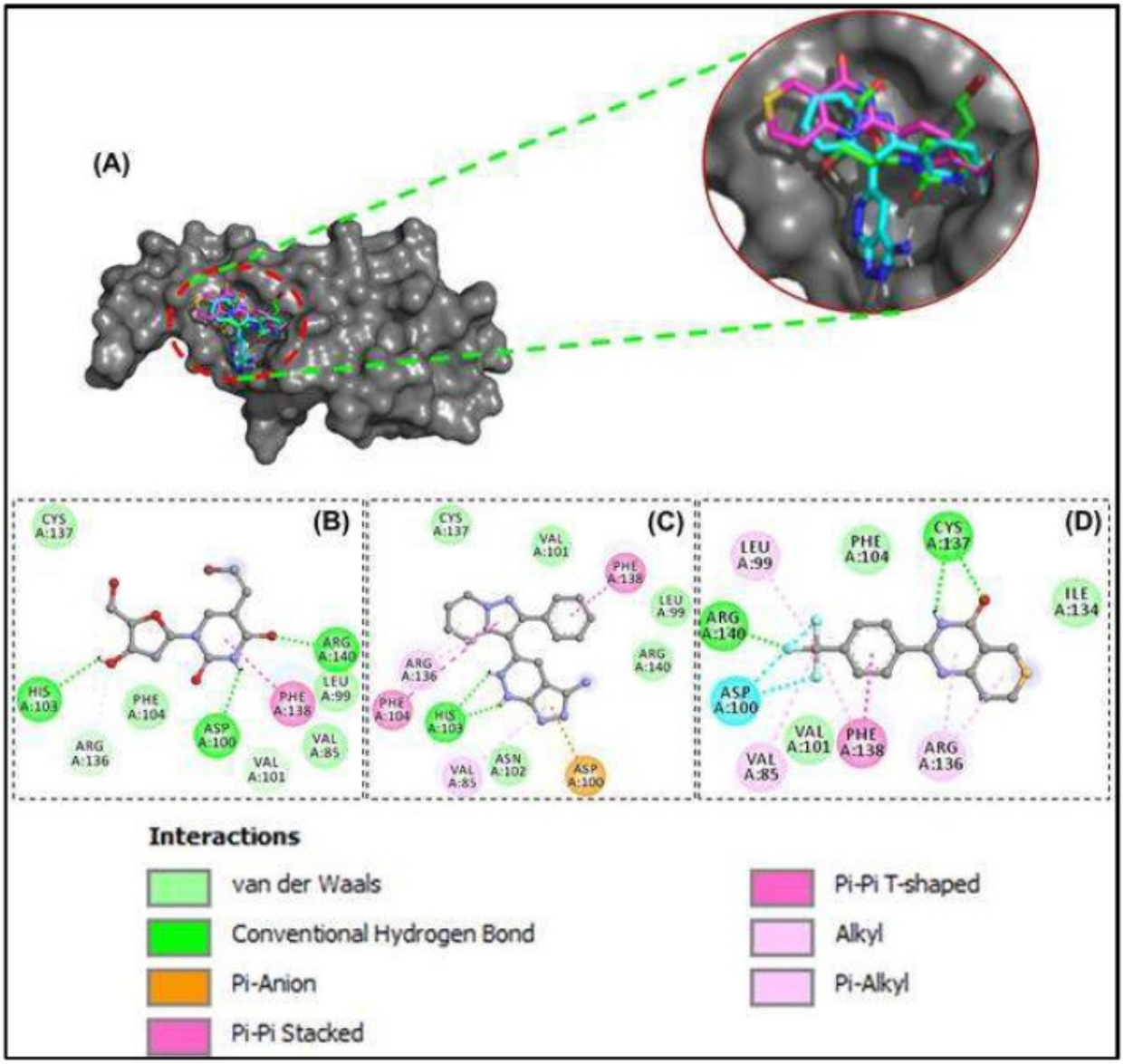
Machine learning and artificial intelligence can be used to assess complex chemical interactions and predict the activity and toxicity of potential drugs in the context of NF-kB-targeted medication discovery and not just in cancer therapy, for example it can be used to repair ischemic stroke injury [125].

3.3.5 Designing HSP Targets

By using computational techniques, small compounds or biologics that can specifically bind to and inhibit HSP activity are found and improved. This process is known as "in silico design of HSP inhibitors."

For example, this [126] study focuses on the potential for creating highly effective, low-toxicity HSP27 inhibitors that are extremely selective, which could circumvent the challenges associated with attacking oncogenic HSP27. Despite the fact that some HSP27 inhibitors were successful in pre-clinical research and clinical trials and despite the fact that HSP27 dysregulation is a feature of many human malignancies, none of them have advanced to the anticancer stage of development. Computer-aided assisted therapeutic discovery and design, an in-silico technique used in this work, entails searching through an anticancer chemical library for new HSP27 inhibitors. The 3D binding arrangement of XAV-939 (pink), FR180204 (cyan), and BVDU (green) is shown in Figure (A). The positions of the three ligands on our protein target were comparable. The molecular interaction fingerprints in 2D demonstrated that our ligands' atoms (B), BVDU, (C), FR180204, and (D), XAV-939 interacted with amino acid residues crucial for

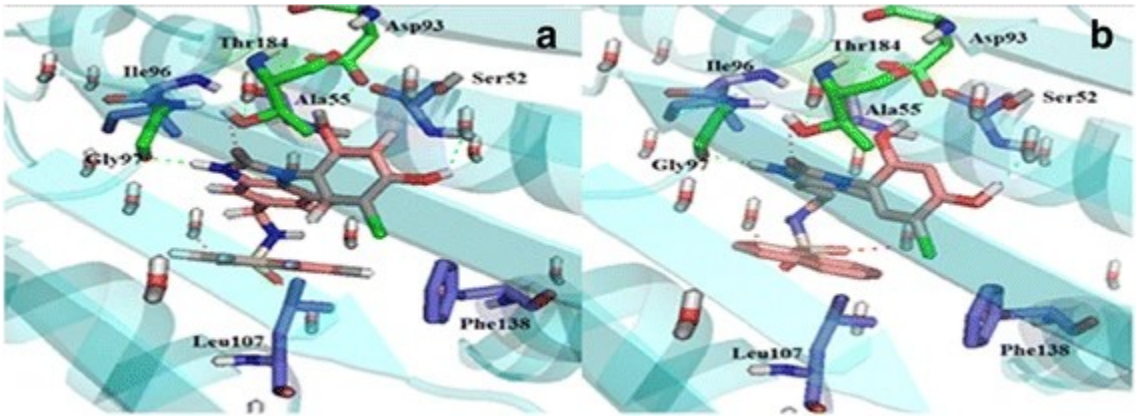
the binding of two HSP27 monomers.



Molecular docking of anti-cancer compounds against HSP27

In another study [127] the team looked at the 3,4-isoxazolediamide derivatives were examined in this work as prospective Hsp90 inhibitors for the treatment of cancer. To predict the inhibitory activity of Hsp90, the researchers used multiple linear regression and genetic algorithms of partial least squares. They validated their models using leave-one-out cross-validation and Y-

randomization tests, and they carried out molecular docking studies to assess the compounds' interactions with Hsp90. The following shows the X-ray crystallography and ligands of the study The crystallography structure 3OWD was used for structure-based drug design, and a docking validation was done by extracting the ligand and redocking it.



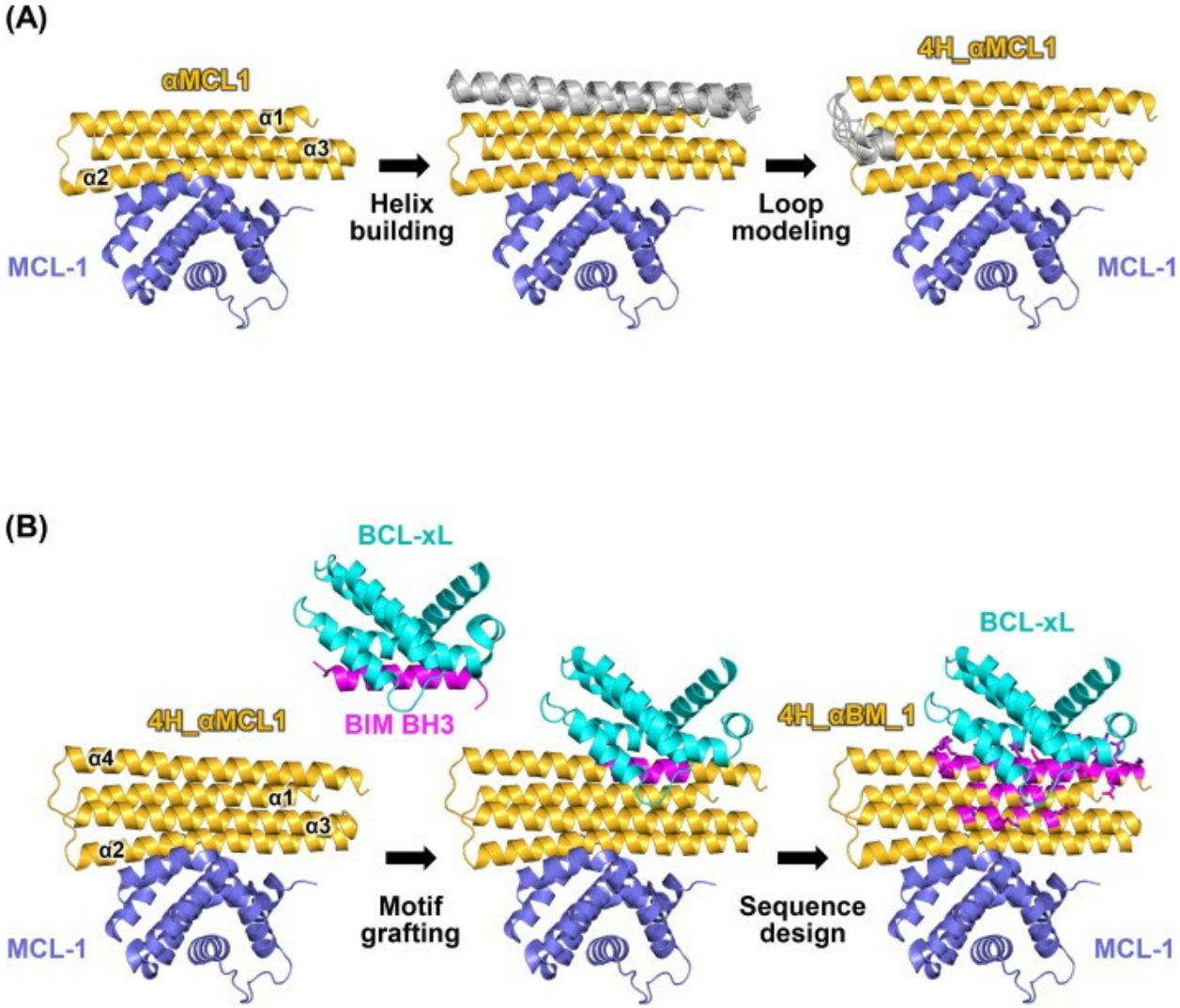
a) The position of ligand in X-ray crystallography, b) The position of ligand after redocking

3.3.6 Designing Bcl-2 Targets

Molecular docking, molecular dynamics simulations, and QSAR (quantitative structure-activity relationship) modeling are in silico techniques that can be utilized for Bcl-2 protein design. These techniques can be used to discover crucial structural elements that influence ligand binding, predict the binding affinity of possible ligands to Bcl-2 proteins, and model the behavior of ligands and proteins over time.

For example, in a 2022 paper [128] This paper describes the computational design of a protein that suppresses concurrently the anti-apoptotic proteins BCL-xL and MCL-1, which are frequently overexpressed in cancer cells. The protein binds with both proteins with high affinity

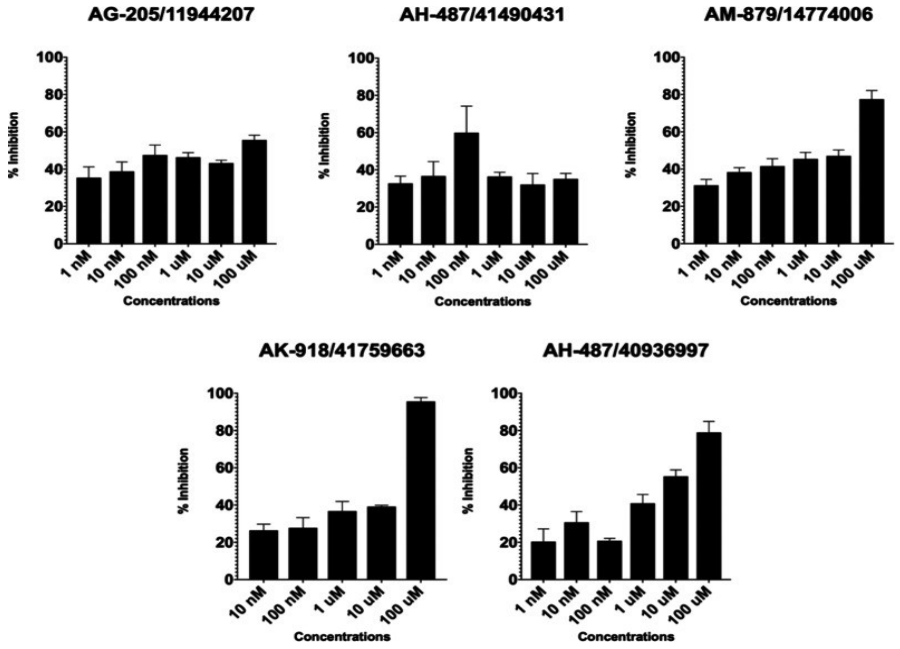
and triggers apoptotic cell death in cancer cells at levels higher than those generated by the pro-apoptotic BIM protein. It was created utilizing a three-helix bundle with a fourth helix added to bind BCL-xL. According to the findings, delivering this dual-binding protein to malignancies specifically may be a useful strategy for inducing cancer cell death.



(A) The computational design procedure used to generate 4H_αMCL1

(B) Motif grafting and sequence design.

In addition to this, in silico has been used to discover novel drugs in treatment. In this paper [129] researchers looked for the BCL-2 (B-Cell Lymphoma 2) proteins, which are important in regulating cell death. The article describes the conflicting functions of proapoptotic and antiapoptotic BCL-2 family members as well as how a cell's expression level can predict its propensity for apoptosis. The study used virtual screening of a library of small molecules to find new BCL-2 inhibitors using molecular modeling and time-resolved fluorescence resonance energy transfer (TR-FRET)-guided screening. The article also discusses the value of computational biology and molecular simulations in drug design. Their TR-Fret analysis shows the inhibitory activity as you can see below.



The *X*-axis shows tested concentrations ranging 1 nM to 100 μM; the *Y*-axis shows the % inhibitory activity relative to samples without inhibitory molecules.

Overall, these computational methods enable the quick screening of sizable compound libraries and the prediction of the binding affinity and potency of possible drug candidates, ultimately resulting in the creation of more efficient and individualized cancer therapies.

3.4 Accuracy and Reliability of Computational Tools

In Silico Design of ACPs relies heavily on computational tools and techniques for predicting the structure, stability, and activity of peptide molecules. Several studies have compared the results of different computational tools and techniques and evaluated their accuracy and reliability. The most effective tools and techniques for predicting peptide activity include:

- Molecular Docking
- Molecular Dynamics Simulation
- Quantitative Structure-Activity Relationship (QSAR)
- Artificial Neural Networks (ANNs)

These tools have been shown to accurately predict the activity of peptide molecules and are therefore recommended for use in In Silico Design of ACPs.

3.5 Strengths and Limitations of In Silico Design of ACPs

Despite its potential, In Silico Design of ACPs has several limitations that must be considered. One common source of variability is the choice of computational tools and techniques used, which can greatly impact the accuracy and reliability of results [130]. Additionally, the accuracy of predictions is highly dependent on the quality and availability of data used to train computational models [131]. Further research is needed to address these limitations and improve the accuracy and reliability of In Silico Design of ACPs.

3.6 Standardized Protocols and Guidelines

The development of more standardized protocols and guidelines for In Silico Design of ACPs is essential for ensuring reproducibility and consistency across studies. Best practices and recommendations for designing and simulating peptides with anticancer activity include:

- Use of multiple computational tools and techniques to validate results [132]
- Use of high-quality data to train computational models [133]
- Use of appropriate statistical and validation methods to assess accuracy and reliability [134]
- Adoption of standardized reporting guidelines for results and methods [135]

The features of AMPs, such as their activity, selectivity, and stability, can be improved by in silico design [136]. The structure of the peptide and its interactions with the target microorganism, for instance, can be predicted using computer simulations, which can then be utilized to develop peptides with increased activity and selectivity. It has been shown to improve survival predictions [137]. It can optimize selective membranolytic ACPs [138]. Not only that, in silico models can be used to design and discover new ACPs [139].

3.7 Existing Literature on In Silico Design of ACPs

The studies covered a range of topics, including the identification of potential targets for peptide-based therapies, the development and validation of computational tools and techniques, and the evaluation of the accuracy and reliability of In Silico Design of ACPs. Overall, the literature

suggests that In Silico Design of ACPs is a promising approach for the treatment of cancer, but further research is needed to address its limitations.

4. DISCUSSION

The findings of this systematic literature review have several implications for future research in In Silico Design of ACPs. One of the key implications is the need for more experimental validation of predicted results. Although computational tools have been shown to be effective in predicting the structure, stability, and activity of peptide molecules, experimental validation is needed to ensure the accuracy and reliability of these predictions. Future research should focus on developing more accurate and reliable experimental methods for validating predicted results.

Another implication of this review is the need for more standardized protocols and guidelines for In Silico Design of ACPs. The lack of standardization in this field has led to variability and bias in the results of different studies. The development of more standardized protocols and guidelines would help to ensure consistency and comparability across studies, and would facilitate the replication and validation of results.

There are several promising peptide-based therapies that we looked at in this study. The first group discussed work by preventing EGFR activation or downstream signaling [140]. EGFR itself is a cell surface protein that binds to the epidermal growth factor [141] and in healthy cells when the epidermal growth factor binds to its receptor, it triggers what we call a series for downstream cell signaling events that eventually result in cellular responses such growth, proliferation, differentiation, and survival [142]. In cancer cells and metastasis this binding and cascade of chemical reactions has become a nuisance. The researchers in the previous section have looked into inhibiting the pathway and this inhibition can have various effects. Examples of peptide-based drugs that target the extracellular domain of EGFR and have acquired FDA approval for the treatment of particular cancers are cetuximab and panitumumab, which are mentioned in the EGFR section of this paper along with Osimertinib. The inhibition of

unchecked cell growth and proliferation, which is frequently observed in cancer cells, is one potential outcome of blocking this mechanism [143]. Angiogenesis, migration, and invasion are other cellular processes that can be impacted by inhibiting the EGF/EGFR pathway and are crucial for the development and spread of tumors [144]. Inhibiting the EGF/EGFR pathway, however, may have negative consequences on healthy cells and tissues as well, as it also plays key functions in normal tissue growth and homeostasis [145].

The second group of peptides that can be used for therapy in cancer treatment is the Vascular Endothelial Growth Factor protein which promotes the formation of new blood vessels, i.e.: angiogenesis [146] but it also promotes wound healing [147], and tumor growth [148]. As we can see in the previous studies mentioned here, the signaling route of VEGF and its relationship with numerous clinical disorders, including cancer, have made them a focus of interest for the creation of innovative therapeutic treatments. Drugs that block the VEGF pathway, such as bevacizumab, that is looked into by the respective researchers, have been created for the treatment of cancer and are used to stop the development of new blood vessels in tumors, reducing or stopping their growth. In fact, bevacizumab, ramucirumab, and aflibercept are VEGF inhibitors that slow the growth and metastasis of cancers by blocking VEGF's action and decreasing the blood flow to them. These drugs target the tumor microenvironment, the area around the tumor that contains various cell types and structures and affects the growth and metastasis of cancer cells. However, we must be reminded that VEGF and its pathway remains important in healing so inhibiting them can have the adverse effect of delayed wound healing [149] which can cause complications post invasive surgery or injuries[150]. There are other potential adverse effects of VEGF inhibitors, such as hypertension [151], proteinuria [152] and thromboembolic events [153].

The third promising peptide-based therapy involves the family protein of Matrix Metalloproteinases which, in moderation, has various uses, such as angiogenesis [154], and tissue remodeling regulation [155]. The problem is when MMPs are overexpressed, they have been associated with several stages of tumor growth and development [156]. The various pathological conditions that ensue from this dysregulation also include arthritis[157] and cardiovascular diseases[158]. It is no wonder that it is a target for the researchers mentioned in this study, however, MMP inhibitors have been shown by our researchers to be toxic and are therefore ineffective in clinical trials.

Meanwhile NF-kB, another group that researchers have taken interest in, is actually functioning as a transcription factor that regulates immune response [159], inflammation [160] and apoptosis[161] As alluded to by the researchers in the earlier paper, inhibiting the activity of NF-kB signaling has been suggested as a potential therapeutic strategy for treating cancer since dysregulation of NF-kB signaling has been associated with several types of cancer [162]. Given this duality of NF-kB regulation, it needs to be logically deduced that blocking NF-kB can also be harmful because it may weaken the immune system and raise the risk of infection. In order to prevent negative effects, the modification of NF-kB activity must be carefully managed.

Heat shock proteins on the other hand, are a group of proteins that are released due to stressors like heat, toxins and radiation [163] but they are very important if well-regulated in terms of protein folding and stability [164]. Like MMPs, cancer results when HSP is overexpressed and conversely, helps the tumor cell survive [165]. The researchers mentioned looked into inhibiting the HSP pathway for example using HSP90. It is important to mention that in cases where the dosage of a therapeutic treatment can change whether the outcome is positive or adverse, that we ensure to find the exact setpoint or set range that does not disturb the delicate balance.

Finally, the Bcl-2 family of proteins controls what is considered the most widely known method of cancer cell treatments, that is apoptosis [166]. The proteins in this family actually control apoptosis in healthy cells too, if not overexpressed. However, much like MMPs and HSPs, if any of the proteins in the Bcl-2 family is over-expressed can lead to resistance to both chemotherapy [167] and radiation treatment [168]. Our researchers have looked into using ABT-199 and ABT-263 inhibitors in order to decrease the activity of these proteins and although their results show some promising results, we need to remember that their analysis is based on preclinical investigations and it is possible that they cannot be replicated in clinical trials.

Since various inhibitors may be more efficient for various types of cancer or for particular stages of cancer, it is impossible to categorically declare one type of inhibitor to be "the best" for the treatment of cancer. Many of the targeted therapies have their own potential promises and the choice of inhibitor is ultimately going to factor in other things like the type of cancer, the stage of cancer and the adverse effects of each therapy.

Finally, in this study, when considering in silico design to develop novel inhibitors of EGF, VEGF, MMP, NF-kB, HSP, or Bcl-2 proteins, researchers used various computational tools and their inhibition. They used techniques for agent design and optimization, including molecular docking, virtual screening, and molecular dynamics simulations. Then researchers validated inhibitor potency through molecular dynamics simulations and free energy calculations, providing insight into the stability and binding affinity of inhibitor-target complexes. Our review suggests that these inhibitors potently inhibit the activity of his EGF, VEGF, MMP, NF-kB, HSP, or Bcl-2 proteins, potentially leading to the induction of apoptosis and inhibition of tumor growth. suggests that there is Using computational tools and techniques, researchers can efficiently identify and optimize potential drug candidates prior to experimental validation. This

approach not only saves time and resources, but also offers a more streamlined and targeted approach to drug development, resulting in improved efficacy and reduced toxicity. the findings of this review highlight the need for more research in several areas, including the development of more accurate and reliable computational tools, the identification of new targets for peptide-based therapies, and the optimization of peptide design and synthesis.

5. CONCLUSION

This study took a deep dive into In Silico Design of ACPs and concluded that it is a promising approach for the treatment of cancer. This systematic literature review provides insights into the most promising targets for peptide-based therapies, the accuracy and reliability of computational tools, the strengths and limitations of In Silico Design of ACPs, standardized protocols and guidelines, existing literature, and gaps in knowledge and future research.

In Silico Design of ACPs is a promising approach for designing and simulating peptide molecules with anticancer activity. This systematic literature review identified several promising targets for peptide-based therapies, evaluated the accuracy and reliability of the computational tools used in In Silico Design of ACPs, provided insights into the strengths and limitations of this approach, informed the development of more standardized protocols and guidelines, synthesized and evaluated the existing literature, and identified gaps in knowledge and priorities for future research efforts.

Limitations

This systematic literature review has several limitations that should be taken into account when interpreting the findings. One limitation is the exclusion of non-English studies, which may have resulted in the exclusion of relevant studies. Another limitation is the quality of the included studies varied, which may have affected the reliability and generalizability of the findings. Finally, not all of the anti-cancer peptides mentioned are FDA approved, some are not even allowed for clinical trials due to high levels of toxicity. Some of those approved by the FDA still have immunogenicity issues. This means that they are not a catch-all treatment plan for cancer,

but it must be mentioned that used in combination, most ACPs have promising prospects in treating a variety of cancers.

6. RECOMMENDATIONS

Based on the findings of this systematic literature review, the following recommendations can be made:

Researchers should continue to develop more accurate and reliable computational tools for In Silico Design of ACPs, taking into account the limitations and challenges of existing methods.

Experimental validation of predicted results should be performed wherever possible to increase the reliability and accuracy of the predictions.

More attention should be paid to the identification of new targets for peptide-based therapies, particularly those that have not yet been explored in depth.

Standardized protocols and guidelines should be developed to ensure consistency and reproducibility across different studies.

Further research should be conducted to optimize peptide design and synthesis, taking into account the unique properties and challenges of ACPs.

Reference:

[1] J. Forbes, "Biochemistry, Peptide," *StatPearls - NCBI Bookshelf*, Aug. 29, 2022. https://www.ncbi.nlm.nih.gov/books/NBK562260/#:~:text=[4]%20The%20fundamental%20nature%20of,biochemical%20process%20in%20the%20body.

[2] "[Insulin effect on metabolism in skeletal muscles and the role of muscles in regulation of glucose homeostasis]," *PubMed*, 2001. https://pubmed.ncbi.nlm.nih.gov/11769387/#:~:text=The%20transport%20across%20certain%20cell,protein%20synthesis%20in%20muscle%20tissue.

[3] "TSH (Thyroid-stimulating hormone) Test." https://medlineplus.gov/lab-tests/tsh-thyroid-stimulating-hormone-test/#:~:text=TSH%20tells%20your%20thyroid%20how,makes%20little%20or%20no%20TSH.

[4] G. S. Raju *et al.*, "Luteinizing hormone and follicle stimulating hormone synergy: A review of role in controlled ovarian hyper-stimulation," *Journal of Human Reproductive Sciences*, vol. 6, no. 4, p. 227, Oct. 2013.

[5] I. Antonowicz, "The Role of Enteropeptidase in the Digestion of Protein and its Development in Human Fetal Small Intestine," *Novartis Foundation Symposium*, pp. 169–187, Jan. 1979.

[6] J. Haginaka, "8.9 Chromatographic Separations and Analysis: Protein and Glycoprotein Stationary Phases," Jan. 2012.

[7] M. R. Sapio and L. D. Fricker, "Carboxypeptidases in disease: Insights from peptidomic studies," *Proteomics Clinical Applications*, vol. 8, no. 5–6, pp. 327–337, Jun. 2014.

[8] M. J. Gilbert and R. E. Pratley, "GLP-1 Analogs and DPP-4 Inhibitors in Type 2 Diabetes Therapy: Review of Head-to-Head Clinical Trials," *Frontiers in Endocrinology*, vol. 11, Apr. 2020.

[9] R. I. Lehrer, C. L. Bevins, and T. Ganz, "Defensins and Other Antimicrobial Peptides and Proteins," *Elsevier eBooks*, pp. 95–110, Jan. 2005.

[10] J. C. Froehlich, "Opioid Peptides," *PubMed Central (PMC)*, 1997. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6826828/

[11] S. J. Arrowsmith and S. Wray, "Oxytocin: Its Mechanism of Action and Receptor Signalling in the Myometrium," *Journal of Neuroendocrinology*, vol. 26, no. 6, pp. 356–369, Jun. 2014.

[12] J. Thundimadathil, "Cancer Treatment Using Peptides: Current Therapies and Future Prospects," *Journal of Amino Acids*, vol. 2012, pp. 1–13, Dec. 2012, doi: 10.1155/2012/967347.

[13] A. K. Tripathi and J. K. Vishwanatha, "Role of Anti-Cancer Peptides as Immunomodulatory Agents: Potential and Design Strategy," *Pharmaceutics*, vol. 14, no. 12, p. 2686, Dec. 2022.

[14] R. J. Boohaker, M. J. Lee, P. Vishnubhotla, J. L. M. Perez, and A. R. Khaled, "The Use of Therapeutic Peptides to Target and to Kill Cancer Cells," *Current Medicinal Chemistry*, vol. 19, no. 22, pp. 3794–3804, Jul. 2012, doi: 10.2174/092986712801661004.

[15] S. Marqus, E. Pirogova, and T. J. Piva, "Evaluation of the use of therapeutic peptides for cancer treatment," *Journal of Biomedical Science*, vol. 24, no. 1, Mar. 2017.

[16] J. Lei, "The antimicrobial peptides and their potential clinical applications," *PubMed Central (PMC)*, 2019. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6684887/#:~:text=Antimicrobial%20peptides%20(AMPs)%20are%20the,the%20action%20of%20cationic%20AMPs.

[17] B. Deslouches and Y. P. Di, "Antimicrobial peptides with selective antitumor mechanisms: prospect for anticancer applications," *Oncotarget*, vol. 8, no. 28, pp. 46635–46651, Mar. 2017.

[18] B. Deslouches and Y. P. Di, "Antimicrobial peptides with selective antitumor mechanisms: prospect for anticancer applications," *Oncotarget*, vol. 8, no. 28, pp. 46635–46651, Mar. 2017.

[19] D. W. Hoskin and A. Ramamoorthy, "Studies on anticancer activities of antimicrobial peptides," *Biochimica Et Biophysica Acta - Biomembranes*, vol. 1778, no. 2, pp. 357–375, Feb. 2008.

[20] K. a M. McMillan and M. R. P. Coombs, "Review: Examining the Natural Role of Amphibian Antimicrobial Peptide Magainin," *Molecules*, vol. 25, no. 22, p. 5436, Nov. 2020.

[21] I. Rady, I. A. Siddiqui, M. Rady, and H. Mukhtar, "Melittin, a major peptide component of bee venom, and its conjugates in cancer therapy," *Cancer Letters*, vol. 402, pp. 16–31, Aug. 2017.

[22] F. Lu, Y. Zhu, G. Zhang, and Z. Liu, "Renovation as innovation: Repurposing human antibacterial peptide LL-37 for cancer therapy," *Frontiers in Pharmacology*, vol. 13, Aug. 2022.

[23] J. Xie *et al.*, "Cell-Penetrating Peptides in Diagnosis and Treatment of Human Diseases: From Preclinical Research to Clinical Application," *Frontiers in Pharmacology*, vol. 11, May 2020.

[24] M. Zhou *et al.*, "The role of cell-penetrating peptides in potential anti-cancer therapy," *Clinical and Translational Medicine*, vol. 12, no. 5, May 2022.

[25] A. Ahsan *et al.*, "Efficacy of an EGFR-Specific Peptide against EGFR-Dependent Cancer Cell Lines and Tumor Xenografts," *Neoplasia*, vol. 16, no. 2, pp. 105-W2, Feb. 2014.

[26] F. A. L. M. Eskens and J. Verweij, "Clinical studies in the development of new anticancer agents exhibiting growth inhibition in models: facing the challenge of a proper study design," *Critical Reviews in Oncology Hematology*, vol. 34, no. 2, pp. 83–88, May 2000.

[27] A. Zambrowicz, M. Timmer, A. Polanowski, G. Lubec, and T. Trziszka, "Manufacturing of peptides exhibiting biological activity," *Amino Acids*, vol. 44, no. 2, pp. 315–320, Feb. 2013.

[28] B. J. Evans, A. J. King, A. Katsifis, L. Matesic, and J. F. Jamie, "Methods to Enhance the Metabolic Stability of Peptide-Based PET Radiopharmaceuticals," *Molecules*, vol. 25, no. 10, p. 2314, May 2020.

[29] R. E. W. Hancock and G. Diamond, "The role of cationic antimicrobial peptides in innate host defences," *Trends in Microbiology*, vol. 8, no. 9, pp. 402–410, Sep. 2000.

[30] N. Sitaram, C. Subbalakshmi, and R. Nagaraj, "Indolicidin, a 13-residue basic antimicrobial peptide rich in tryptophan and proline, interacts with Ca2+-calmodulin," *Biochemical and Biophysical Research Communications*, vol. 309, no. 4, pp. 879–884, Oct. 2003.

[31] L.-Y. Huang *et al.*, "Dermaseptin-PH: A Novel Peptide with Antimicrobial and Anticancer Activities from the Skin Secretion of the South American Orange-Legged Leaf Frog, Pithecopus (Phyllomedusa) hypochondrialis," *Molecules*, vol. 22, no. 10, p. 1805, Oct. 2017.

[32] K. a M. McMillan and M. R. P. Coombs, "Review: Examining the Natural Role of Amphibian Antimicrobial Peptide Magainin," *Molecules*, vol. 25, no. 22, p. 5436, Nov. 2020.

[33] "The anti-cancer peptide, PNC-27, induces tumor cell necrosis of a poorly differentiated non-solid tissue human leukemia cell line that depends on expression of HDM-2 in the plasma membrane of these cells," *PubMed*. https://pubmed.ncbi.nlm.nih.gov/25117093/

[34] M. Liu, J. Lv, L. Chen, W. Li, and W. Han, "In Silico Discovery of Anticancer Peptides from Sanghuang," *International Journal of Molecular Sciences*, vol. 23, no. 22, p. 13682, Nov. 2022.

[35] N. Schaduangrat, C. Nantasenamat, V. Prachayasittikul, and W. Shoombuatong, "ACPred: A Computational Tool for the Prediction and Analysis of Anticancer Peptides," *Molecules*, vol. 24, no. 10, p. 1973, May 2019.

[36] L. R. Hart, C. G. Lebedenko, S. M. Mitchell, R. E. Daso, and I. A. Banerjee, "In Silico Studies of Tumor Targeted Peptide-Conjugated Natural Products for Targeting Over-Expressed Receptors in Breast Cancer Cells Using Molecular Docking, Molecular Dynamics and MMGBSA Calculations," *Applied Sciences*, vol. 12, no. 1, p. 515, Jan. 2022.

[37] R. Thevendran, T. Tang, and M. Citartan, "*In-silico* selection employing rigid docking and molecular dynamic simulation in selecting DNA aptamers against androgen receptor," *Biotechnology Journal*, p. 2200092, Feb. 2023.

[38] Z. Hashemi *et al.*, "In silico Approaches for the Design and Optimization of Interfering Peptides Against Protein–Protein Interactions," *Frontiers in Molecular Biosciences*, vol. 8, Apr. 2021.

[39] M. K. Fath *et al.*, "Anti-cancer peptide-based therapeutic strategies in solid tumors," *Cellular & Molecular Biology Letters*, vol. 27, no. 1, Apr. 2022.

[40] M. Arruebo *et al.*, "Assessment of the Evolution of Cancer Treatment Therapies," *Cancers*, vol. 3, no. 3, pp. 3279–3330, Aug. 2011.

[41] E. Sarafraz-Yazdi, "Ex vivo Efficacy of Anti-Cancer Drug PNC-27 in the Treatment of Patient-Derived Epithelial Ovarian Cancer," Nov. 01, 2015. http://www.annclinlabsci.org/content/45/6/650.long

[42] E. Pasquereau-Kotula, J. Habault, G. Kroemer, and J.-L. Poyet, "The anticancer peptide RT53 induces immunogenic cell death," *PLOS ONE*, vol. 13, no. 8, p. e0201220, Aug. 2018.

[43] A. M. Alqarni, V. A. Ferro, J. Parkinson, M. J. Dufton, and D. I. Watson, "Effect of Melittin on Metabolomic Profile and Cytokine Production in PMA-Differentiated THP-1 Cells," *Vaccines*, vol. 6, no. 4, p. 72, Oct. 2018.

[44] E. G. Findlay *et al.*, "Exposure to the antimicrobial peptide LL-37 produces dendritic cells optimized for immunotherapy," *OncoImmunology*, vol. 8, no. 8, May 2019.

[45] A. M. Taylor, "Alpha-Melanocyte Stimulating Hormone (α-MSH) Is a Post-Caspase Suppressor of Apoptosis in RAW 264.7 Macrophages," *PLOS ONE*, vol. 8, no. 8, p. e74488, Aug. 2013.

[46] J. K. Actor, S.-A. Hwang, and M. L. Kruzel, "Lactoferrin as a Natural Immune Modulator," *Current Pharmaceutical Design*, vol. 15, no. 17, pp. 1956–1973, May 2009.

[47] A. Gokhale and S. D. Satyanarayanajois, "Peptides and peptidomimetics as immunomodulators," *Immunotherapy*, vol. 6, no. 6, pp. 755–774, Jul. 2014.

[48] L. Wang *et al.*, "Therapeutic peptides: current applications and future directions," *Signal Transduction and Targeted Therapy*, vol. 7, no. 1, Feb. 2022.

[49] L. Yao *et al.*, "Accelerating the Discovery of Anticancer Peptides through Deep Forest Architecture with Deep Graphical Representation," *International Journal of Molecular Sciences*, vol. 24, no. 5, p. 4328, Feb. 2023.

[50] S. Gupta, P. Kapoor, K. Chaudhary, A. Gautam, R. Kumar, and G. P. S. Raghava, "In Silico Approach for Predicting Toxicity of Peptides and Proteins," *PLOS ONE*, vol. 8, no. 9, p. e73957, Sep. 2013.

[51] J. H. Hwang, S.-G. Kim, T. J. Shin, Y. H. Jang, H. J. Kwon, and G. Lee, "Development of Anticancer Peptides Using Artificial Intelligence and Combinational Therapy for Cancer Therapeutics," *Pharmaceutics*, vol. 14, no. 5, p. 997, May 2022.

[52] A. Tyagi, P. Kapoor, R. Kumar, K. Chaudhary, A. Gautam, and G. P. S. Raghava, "In Silico Models for Designing and Discovering Novel Anticancer Peptides," *Scientific Reports*, vol. 3, no. 1, Oct. 2013.

[53] S. Basith, B. Manavalan, T. J. Shin, D. Y. Lee, and G. Lee, "Evolution of Machine Learning Algorithms in the Prediction and Design of Anticancer Peptides," *Current Protein & Peptide Science*, vol. 21, no. 12, pp. 1242–1250, Jan. 2020.

[54] Q. Wu, H. Ke, D. Li, Q. Wang, J. Fang, and Z. Jingwei, "Recent Progress in Machine Learning-based Prediction of Peptide Activity for Drug Discovery," *Current Topics in Medicinal Chemistry*, vol. 19, no. 1, pp. 4–16, Jan. 2019.

[55] B. Brooks *et al.*, "CHARMM: The biomolecular simulation program," *Journal of Computational Chemistry*, vol. 30, no. 10, pp. 1545–1614, Jul. 2009.

[56] M. S. Pacella, D. S. Koo, R. A. Thottungal, and J. J. Gray, "Using the RosettaSurface Algorithm to Predict Protein Structure at Mineral Surfaces," *Methods in Enzymology*, pp. 343–366, Jan. 2013.

[57] A. Sinelnikova and D. Van Der Spoel, "NMR refinement and peptide folding using the GROMACS software," *Journal of Biomolecular NMR*, vol. 75, no. 4–5, pp. 143–149, Mar. 2021.

[58] M. J. Thun, J. O. L. DeLancey, A. Jemal, and E. C. Ward, "The global burden of cancer: priorities for prevention," *Carcinogenesis*, vol. 31, no. 1, pp. 100–110, Jan. 2010.

[59] Y. Chen *et al.*, "TPGS-1000 exhibits potent anticancer activity for hepatocellular carcinoma in vitro and in vivo," *Aging*, vol. 12, no. 2, pp. 1624–1642, Jan. 2020.

[60] "The anti-cancer peptide, PNC-27, induces tumor cell necrosis of a poorly differentiated non-solid tissue human leukemia cell line that depends on expression of HDM-2 in the plasma membrane of these cells," *PubMed*. https://pubmed.ncbi.nlm.nih.gov/25117093/

[61] "Anti-Cancer Tumor Cell Necrosis of Epithelial Ovarian Cancer Cell Lines Depends on High Expression of HDM-2 Protein in Their Membranes," *PubMed*, Sep. 01, 2020. https://pubmed.ncbi.nlm.nih.gov/33067207

[62] W. B. Bowne *et al.*, "The Penetratin Sequence in the Anticancer PNC-28 Peptide Causes Tumor Cell Necrosis Rather Than Apoptosis of Human Pancreatic Cancer Cells," *Annals of Surgical Oncology*, vol. 15, no. 12, pp. 3588–3600, Oct. 2008.

[63] V. Adler *et al.*, "Two peptides derived from ras-p21 induce either phenotypic reversion or tumor cell necrosis of ras-transformed human cancer cells," *Cancer Chemotherapy and Pharmacology*, Aug. 2008.

[64] A. J. Berdis, "Inhibiting DNA Polymerases as a Therapeutic Intervention against Cancer," *Frontiers in Molecular Biosciences*, vol. 4, Nov. 2017.

[65] M. B. Ujiki *et al.*, "Apigenin inhibits pancreatic cancer cell proliferation through G2/M cell cycle arrest," *Molecular Cancer*, vol. 5, no. 1, Dec. 2006.

[66] D. H. Koo, T. N. Sut, S. Y. Tan, B. H. Yoon, and J. A. Jackman, "Biophysical Characterization of LTX-315 Anticancer Peptide Interactions with Model Membrane Platforms: Effect of Membrane Surface Charge," *International Journal of Molecular Sciences*, vol. 23, no. 18, p. 10558, Sep. 2022.

[67] A. Yaghoubi, M. Ghayour-Mobarhan, S. M. Hasanian, W. C. Cho, and S. Soleimanpour, “p28 Bacterial Peptide, as an Anticancer Agent,” *Frontiers in Oncology*, vol. 10, Aug. 2020,.

[68] C. Huang, F. Liu, Q. Feng, X.-Y. Pan, S. Song, and J.-L. Yang, “RGD4C peptide mediates anti-p21Ras scFv entry into tumor cells and produces an inhibitory effect on the human colon cancer cell line SW480,” *BMC Cancer*, vol. 21, no. 1, Mar. 2021.

[69] P. Wee and Z. Wang, “Epidermal Growth Factor Receptor Cell Proliferation Signaling Pathways,” *Cancers*, vol. 9, no. 5, p. 52, May 2017.

[70] B. R. Voldborg, L. Damstrup, M. Spang-Thomsen, and H. F. Poulsen, “Epidermal growth factor receptor (EGFR) and EGFR mutations, function and possible role in clinical trials,” *Annals of Oncology*, vol. 8, no. 12, pp. 1197–1206, Dec. 1997.

[71] D. E. Gerber, “EGFR inhibition in the treatment of non-small cell lung cancer,” *Drug Development Research*, vol. 69, no. 6, pp. 359–372, Sep. 2008.

[72] B. Janani *et al.*, “EGFR-Based Targeted Therapy for Colorectal Cancer—Promises and Challenges,” *Vaccines*, vol. 10, no. 4, p. 499, Mar. 2022.

[73] H. Rehmani and N. Issaeva, “EGFR in head and neck squamous cell carcinoma: exploring possibilities of novel drug combinations,” *Annals of Translational Medicine*, vol. 8, no. 13, p. 813, Apr. 2020.

[74] C. Huang, F. Liu, Q. Feng, X.-Y. Pan, S. Song, and J.-L. Yang, “RGD4C peptide mediates anti-p21Ras scFv entry into tumor cells and produces an inhibitory effect on the human colon cancer cell line SW480,” *BMC Cancer*, vol. 21, no. 1, Mar. 2021.

[75] B. Burtness, M. A. Goldwasser, W. A. Flood, B. I. Mattar, and A. A. Forastiere, “Phase III Randomized Trial of Cisplatin Plus Placebo Compared With Cisplatin Plus Cetuximab in Metastatic/Recurrent Head and Neck Cancer: An Eastern Cooperative Oncology Group Study,” *Journal of Clinical Oncology*, vol. 23, no. 34, pp. 8646–8654, Dec. 2005.

[76] J. B. Vermorken *et al.*, “Platinum-Based Chemotherapy plus Cetuximab in Head and Neck Cancer,” *The New England Journal of Medicine*, vol. 359, no. 11, pp. 1116–1127, Sep. 2008.

[77] F. Battaglin, A. Puccini, S. A. Djaballah, and H.-J. Lenz, “<p>The impact of panitumumab treatment on survival and quality of life in patients with <em>RAS</em> wild-type metastatic colorectal cancer</p>,” *Cancer Management and Research*, vol. Volume 11, pp. 5911–5924, Jun. 2019.

[78] J. Vansteenkiste, “Gefitinib (Iressa®): a novel treatment for non-small cell lung cancer,” *Expert Review of Anticancer Therapy*, vol. 4, no. 1, pp. 5–17, Feb. 2004.

[79] C. M. Rocha-Lima, “Erlotinib (Tarceva) for the Treatment of Non–Small-Cell Lung Cancer and Pancreatic Cancer,” *PubMed Central (PMC)*, Oct. 01, 2009. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2799146/

[80] E. D. Deeks and G. M. Keating, "Afatinib in advanced NSCLC: a profile of its use," *Drugs & Therapy Perspectives*, Feb. 2018.

[81] "Osimertinib in Advanced Lung Cancer with EGFR Mutations," *National Cancer Institute*, Dec. 12, 2019. https://www.cancer.gov/news-events/cancer-currents-blog/2019/osimertinib-lung-cancer-improves-survival-flaura

[82] L. J. Scott, "Osimertinib as first-line therapy in advanced NSCLC: a profile of its use," *Drugs & Therapy Perspectives*, vol. 34, no. 8, pp. 351–357, Jul. 2018.

[83] J. Garcia *et al.*, "Bevacizumab (Avastin®) in cancer treatment: A review of 15 years of clinical experience and future outlook," *Cancer Treatment Reviews*, vol. 86, p. 102017, Jun. 2020.

[84] K. S. Tewari *et al.*, "Improved Survival with Bevacizumab in Advanced Cervical Cancer," *The New England Journal of Medicine*, vol. 370, no. 8, pp. 734–743, Feb. 2014.

[85] O. Arrieta, Z. L. Zatarain-Barrón, A. F. Cardona, A. Carmona, and M. Lopez-Mejia, "Ramucirumab in the treatment of non-small cell lung cancer," *Expert Opinion on Drug Safety*, vol. 16, no. 5, pp. 637–644, Apr. 2017.

[86] F. L. Muñoz *et al.*, "Ramucirumab use in patients with advanced gastric cancer (AGC) or gastro-oesophageal junction (GEJ) adenocarcinoma in Spain: RAMIS observational study," *Annals of Oncology*, Oct. 2019.

[87] U. Khan and M. A. Shah, "Ramucirumab for the treatment of gastric or gastro-esophageal junction cancer," *Expert Opinion on Biological Therapy*, vol. 19, no. 11, pp. 1135–1141, Aug. 2019.

[88] M. Jung *et al.*, "Efficacy and tolerability of ramucirumab monotherapy or in combination with paclitaxel in gastric cancer patients from the Expanded Access Program Cohort by the Korean Cancer Study Group (KCSG)," *Gastric Cancer*, vol. 21, no. 5, pp. 819–830, Sep. 2018.

[89] J. W. Moroney, A. K. Sood, and R. L. Coleman, "Aflibercept in epithelial ovarian carcinoma," *Future Oncology*, vol. 5, no. 5, pp. 591–600, Jun. 2009.

[90] J. W. Neal and H. A. Wakelee, "Aflibercept in lung cancer," *Expert Opinion on Biological Therapy*, vol. 13, no. 1, pp. 115–120, Jan. 2013.

[91] P. A. Tang and M. A. S. Moore, "Aflibercept in the treatment of patients with metastatic colorectal cancer: latest findings and interpretations," *Therapeutic Advances in Gastroenterology*, vol. 6, no. 6, pp. 459–473, Sep. 2013.

[92] A. Redondo, B. Castelo, A. Pinto, P. Zamora, and E. Espinosa, "Prolonged Response to Aflibercept in Ovarian Cancer Relapse: A Case Report," *Tumori Journal*, Mar. 2015.

[93] W. P. Tew *et al.*, "Intravenous aflibercept in patients with platinum-resistant, advanced ovarian cancer: Results of a randomized, double-blind, phase 2, parallel-arm study," *Cancer*, vol. 120, no. 3, pp. 335–343, Feb. 2014.

[94] R. P. Verma and C. Hansch, “Matrix metalloproteinases (MMPs): Chemical–biological functions and (Q)SARs,” *Bioorganic & Medicinal Chemistry*, vol. 15, no. 6, pp. 2223–2268, Mar. 2007.

[95] H. W. Hirte *et al.*, “A phase III randomized trial of BAY 12-9566 (tanomastat) as maintenance therapy in patients with advanced ovarian cancer responsive to primary surgery and paclitaxel/platinum containing chemotherapy: A National Cancer Institute of Canada Clinical Trials Group Study,” *Gynecologic Oncology*, vol. 102, no. 2, pp. 300–308, Aug. 2006.

[96] A. M. Winer, S. Adams, and P. Mignatti, “Matrix Metalloproteinase Inhibitors in Cancer Therapy: Turning Past Failures Into Future Successes,” *Molecular Cancer Therapeutics*, vol. 17, no. 6, pp. 1147–1155, Jun. 201.

[97] A. Oeckinghaus and S. Ghosh, “The NF- B Family of Transcription Factors and Its Regulation,” *Cold Spring Harbor Perspectives in Biology*, vol. 1, no. 4, p. a000034, Oct. 2009.

[98] W. E. Naugler and M. Karin, “NF-κB and cancer—identifying targets and mechanisms,” *Current Opinion in Genetics & Development*, vol. 18, no. 1, pp. 19–26, Feb. 2008.

[99] W. Wang, S. Nag, and R. Zhang, “Targeting the NFκB Signaling Pathways for Breast Cancer Prevention and Therapy,” *Current Medicinal Chemistry*, vol. 22, no. 2, pp. 264–289, Dec. 2014.

[100] J. Zhou, H. Zhang, P. Gu, J. Bai, J. B. Margolick, and Y. Zhang, “NF-κB pathway inhibitors preferentially inhibit breast cancer stem-like cells,” *Breast Cancer Research and Treatment*, vol. 111, no. 3, pp. 419–427, Oct. 2008.

[101] A. Steptoe and L. Brydon, “Psychosocial Factors and Coronary Heart Disease: The Role of Psychoneuroimmunological Processes,” *Elsevier eBooks*, pp. 945–974, Dec. 2007.

[102] S. K. Calderwood and J. Gong, “Heat Shock Proteins Promote Cancer: It’s a Protection Racket,” *Trends in Biochemical Sciences*, vol. 41, no. 4, pp. 311–323, Apr. 2016.

[103] H. Lee *et al.*, “Unleashing the full potential of Hsp90 inhibitors as cancer therapeutics through simultaneous inactivation of Hsp90, Grp94, and TRAP1,” *Experimental and Molecular Medicine*, vol. 52, no. 1, pp. 79–91, Jan. 2020.

[104] W. Wen, W. Liu, Y. Shao, and L. Chen, “VER-155008, a small molecule inhibitor of HSP70 with potent anti-cancer activity on lung cancer cell lines,” *Experimental Biology and Medicine*, vol. 239, no. 5, pp. 638–645, Mar. 2014.

[105] K. Kobayashi *et al.*, “Heat shock protein 70 positively regulates transforming growth factor-α-induced hepatocellular carcinoma cell migration via the AKT signaling pathway,” *Heliyon*, vol. 6, no. 9, p. e05002, Sep. 2020.

[106] Z.-N. Li and Y. Luo, “HSP90 inhibitors and cancer: Prospects for use in targeted therapies (Review),” *Oncology Reports*, vol. 49, no. 1, Nov. 2022.

[107] K. Jhaveri, T. Taldone, S. Modi, and G. Chiosis, "Advances in the clinical development of heat shock protein 90 (Hsp90) inhibitors in cancers," *Biochimica Et Biophysica Acta. Molecular Cell Research*, vol. 1823, no. 3, pp. 742–755, Mar. 2012.

[108] J. M. Hardwick and L. Soane, "Multiple Functions of BCL-2 Family Proteins," *Cold Spring Harbor Perspectives in Biology*, vol. 5, no. 2, p. a008722, Feb. 2013.

[109] S. Thomas *et al.*, "Targeting the Bcl-2 family for cancer therapy," *Expert Opinion on Therapeutic Targets*, vol. 17, no. 1, pp. 61–75, Jan. 2013.

[110] E. Valentini *et al.*, "Targeting the anti-apoptotic Bcl-2 family proteins: machine learning virtual screening and biological evaluation of new small molecules," *Theranostics*, vol. 12, no. 5, pp. 2427–2444, Jan. 2022.

[111] G. Itchaki and J. R. Brown, "The potential of venetoclax (ABT-199) in chronic lymphocytic leukemia," *Therapeutic Advances in Hematology*, vol. 7, no. 5, pp. 270–287, Jul. 2016.

[112] A. M. Lopez *et al.*, "Co-targeting of BAX and BCL-XL proteins broadly overcomes resistance to apoptosis in cancer," *Nature Communications*, vol. 13, no. 1, Mar. 2022.

[113] M. P. Goetz, M. S. Hoetker, M. Diken, P. Galle, and R. Kiesslich, "In vivo molecular imaging with cetuximab, an anti-EGFR antibody, for prediction of response in xenograft models of human colorectal cancer," *Endoscopy*, vol. 45, no. 06, pp. 469–477, Apr. 2013, doi: 10.1055/s-0032-1326361.

[114] A. R. Tan *et al.*, "Pharmacokinetics of Cetuximab After Administration of Escalating Single Dosing and Weekly Fixed Dosing in Patients with Solid Tumors," *Clinical Cancer Research*, vol. 12, no. 21, pp. 6517–6522, Nov. 2006, doi: 10.1158/1078-0432.ccr-06-0705.

[115] S. L. M. Gargari and R. R. Adriani, "In Silico Designing and Optimization of the Anti-EGFR Scaffolds by CDRs-Grafting Technique," *Authorea (Authorea)*, Dec. 2022, doi: 10.22541/au.166999155.58450730/v1.

[116] A. Sadremomtaz *et al.*, "Molecular docking, synthesis and biological evaluation of Vascular Endothelial Growth Factor (VEGF) B based peptide as antiangiogenic agent targeting the second domain of the Vascular Endothelial Growth Factor Receptor 1 (VEGFR1D2) for anticancer application," *Signal Transduction and Targeted Therapy*, vol. 5, no. 1, Jun. 2020, doi: 10.1038/s41392-020-0177-z.

[117] A. Salimi, J. H. Lim, J. H. Jang, and J. Y. Lee, "The use of machine learning modeling, virtual screening, molecular docking, and molecular dynamics simulations to identify potential VEGFR2 kinase inhibitors," *Scientific Reports*, vol. 12, no. 1, Nov. 2022, doi: 10.1038/s41598-022-22992-6.

[118] E. Rathi, A. Kumar, and S. G. Kini, "Molecular dynamics guided insight, binding free energy calculations and pharmacophore-based virtual screening for the identification of potential VEGFR2 inhibitors," *Journal of Receptors and Signal Transduction*, vol. 39, no. 5–6, pp. 415–433, Nov. 2019, doi: 10.1080/10799893.2019.1690509.

[119] D. Rathee, V. Lather, and H. Dureja, “Pharmacophore modeling and 3D QSAR studies for prediction of matrix metalloproteinases inhibitory activity of hydroxamate derivatives,” *Biotechnology Research and Innovation*, vol. 1, no. 1, pp. 112–122, Jan. 2017, doi: 10.1016/j.biori.2017.10.002.

[120] A. Ahmad, “Molecular docking and inhibition of matrix metalloproteinase-2 by novel difluorinatedbenzylidene curcumin analog,” *PubMed Central (PMC)*, 2015. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4399093/

[121] Q. Hu and M. Ecker, “Overview of MMP-13 as a Promising Target for the Treatment of Osteoarthritis,” *International Journal of Molecular Sciences*, vol. 22, no. 4, p. 1742, Feb. 2021, doi: 10.3390/ijms22041742.

[122] D. Rathee, V. Lather, and H. Dureja, “Pharmacophore modeling and 3D QSAR studies for prediction of matrix metalloproteinases inhibitory activity of hydroxamate derivatives,” *Biotechnology Research and Innovation*, vol. 1, no. 1, pp. 112–122, Jan. 2017, doi: 10.1016/j.biori.2017.10.002.

[123] V. Mukund, S. K. Behera, A. Alam, and G. P. Nagaraju, “Molecular docking analysis of nuclear factor-κB and genistein interaction in the context of breast cancer,” *Bioinformation*, vol. 15, no. 1, pp. 11–17, Jan. 2019, doi: 10.6026/97320630015011.

[124] T. Kanan, D. Kanan, I. Erol, S. Yazdi, M. Stein, and S. Durdagi, “Targeting the NF-κB/IκBα complex via fragment-based E-Pharmacophore virtual screening and binary QSAR models,” *Journal of Molecular Graphics & Modelling*, vol. 86, pp. 264–277, Jan. 2019, doi: 10.1016/j.jmgm.2018.09.014.

[125] N. Wang *et al.*, “Machine learning enables discovery of Gentianine targeting TLR4/NF-κB pathway to repair ischemic stroke injury,” *Pharmacological Research*, vol. 173, p. 105913, Nov. 2021, doi: 10.1016/j.phrs.2021.105913.

[126] H. I. Umar *et al.*, “Discovery of Novel HSP27 Inhibitors as Prospective Anti-Cancer Agents Utilizing Computer-Assisted Therapeutic Discovery Approaches,” *Cells*, vol. 11, no. 15, p. 2412, Aug. 2022, doi: 10.3390/cells11152412.

[127] M. Abbasi, H. Sadeghi-Aliabadi, and M. Amanlou, “Prediction of new Hsp90 inhibitors based on 3,4-isoxazolediamide scaffold using QSAR study, molecular docking and molecular dynamic simulation,” *DARU*, vol. 25, no. 1, Jun. 2017, doi: 10.1186/s40199-017-0182-0.

[128] S. H. Kim, H. Park, and B.-H. Oh, “Computational design of an apoptogenic protein that binds BCL-xL and MCL-1 simultaneously and potently,” *Computational and Structural Biotechnology Journal*, vol. 20, pp. 3019–3029, Jun. 2022, doi: 10.1016/j.csbj.2022.06.021.

[129] K. Sahin, M. D. Orhan, T. Avsar, and S. Durdagi, “Hybrid *In Silico* and TR-FRET-Guided Discovery of Novel BCL-2 Inhibitors,” *ACS Pharmacology & Translational Science*, vol. 4, no. 3, pp. 1111–1123, Apr. 2021, doi: 10.1021/acsptsci.0c00210.

[130] W. Shoombuatong, N. Schaduangrat, and C. Nantasenamat, “Unraveling the bioactivity of anticancer peptides as deduced from machine learning.,” *Excli Journal*, vol. 17, pp. 734–752, Jul. 2018, doi: 10.17179/excli2018-1447.

[131] F. E. Agamah *et al.*, “Computational/in silico methods in drug target and lead prediction,” *Briefings in Bioinformatics*, vol. 21, no. 5, pp. 1663–1675, Sep. 2020, doi: 10.1093/bib/bbz103.

[132] G. Hessler and K.-H. Baringhaus, “Artificial Intelligence in Drug Design,” *Molecules*, vol. 23, no. 10, p. 2520, Oct. 2018, doi: 10.3390/molecules23102520.

[133] C. B. Collin *et al.*, “Computational Models for Clinical Applications in Personalized Medicine—Guidelines and Recommendations for Data Integration and Model Validation,” *Journal of Personalized Medicine*, vol. 12, no. 2, p. 166, Jan. 2022, doi: 10.3390/jpm12020166.

[134] M. Viceconti, F. Pappalardo, B. Rodriguez, M. Horner, J. E. Bischoff, and F. M. Tshinanu, “In silico trials: Verification, validation and uncertainty quantification of predictive models used in the regulatory evaluation of biomedical products,” *Methods*, vol. 185, pp. 120–127, Jan. 2021, doi: 10.1016/j.ymeth.2020.01.011.

[135] S. Brunak *et al.*, “Towards standardization guidelines for *in silico* approaches in personalized medicine,” *Journal of Integrative Bioinformatics*, vol. 17, no. 2–3, Jul. 2020, doi: 10.1515/jib-2020-0006.

[136] R. Ghosh, N. Oak, and S. E. Plon, “Evaluation of in silico algorithms for use with ACMG/AMP clinical variant interpretation guidelines,” *Genome Biology*, vol. 18, no. 1, Nov. 2017, doi: 10.1186/s13059-017-1353-5.

[137] C.-J. Lee *et al.*, “Machine learning with in silico analysis markedly improves survival prediction modeling in colon cancer patients,” *Cancer Medicine*, vol. 12, no. 6, pp. 7603–7615, Nov. 2022, doi: 10.1002/cam4.5420.

[138] G. Gabernet *et al.*, “In silico design and optimization of selective membranolytic anticancer peptides,” *Scientific Reports*, vol. 9, no. 1, Aug. 2019, doi: 10.1038/s41598-019-47568-9.

[139] A. Tyagi, P. Kapoor, R. Kumar, K. Chaudhary, A. Gautam, and G. P. S. Raghava, “In Silico Models for Designing and Discovering Novel Anticancer Peptides,” *Scientific Reports*, vol. 3, no. 1, Oct. 2013, doi: 10.1038/srep02984.

[140] E. Hopper-Borge, R. E. Nasto, V. Ratushny, L. M. Weiner, E. A. Golemis, and I. Astsaturov, “Mechanisms of tumor resistance to EGFR-targeted therapies,” *Expert Opinion on Therapeutic Targets*, vol. 13, no. 3, pp. 339–362, May 2009, doi: 10.1517/14712590902735795.

[141] “EGFR epidermal growth factor receptor [Homo sapiens (human)] - Gene - NCBI.” https://www.ncbi.nlm.nih.gov/gene/1956#:~:text=EGFR%20is%20a%20cell%20surface,autophosphorylation%20leading%20to%20cell%20proliferation.

[142] P. Wee and Z. Wang, “Epidermal Growth Factor Receptor Cell Proliferation Signaling Pathways,” *Cancers*, vol. 9, no. 5, p. 52, May 2017, doi: 10.3390/cancers9050052.

[143] G. M. Cooper, “The Development and Causes of Cancer,” *The Cell - NCBI Bookshelf*, 2000. https://www.ncbi.nlm.nih.gov/books/NBK9963/

[144] R. Lugano, M. Ramachandran, and A. Dimberg, “Tumor angiogenesis: causes, consequences, challenges and opportunities,” *Cellular and Molecular Life Sciences*, vol. 77, no. 9, pp. 1745–1770, May 2020, doi: 10.1007/s00018-019-03351-7.

[145] V. Hirsh, “Managing Treatment-Related Adverse Events Associated with egfr Tyrosine Kinase Inhibitors in Advanced Non-Small-Cell Lung Cancer,” *Current Oncology*, vol. 18, no. 3, pp. 126–138, Jun. 2011, doi: 10.3747/co.v18i3.877.

[146] J. J. Haigh, “Role of VEGF in organogenesis,” *Organogenesis*, vol. 4, no. 4, pp. 247–256, Oct. 2008, doi: 10.4161/org.4.4.7415.

[147] K. E. Johnson and T. A. Wilgus, “Vascular Endothelial Growth Factor and Angiogenesis in the Regulation of Cutaneous Wound Repair,” *Advances in Wound Care*, vol. 3, no. 10, pp. 647–661, Oct. 2014, doi: 10.1089/wound.2013.0517.

[148] I. Bekes *et al.*, “Does VEGF facilitate local tumor growth and spread into the abdominal cavity by suppressing endothelial cell adhesion, thus increasing vascular peritoneal permeability followed by ascites production in ovarian cancer?,” *Molecular Cancer*, vol. 15, no. 1, Feb. 2016, doi: 10.1186/s12943-016-0497-3.

[149] T. Kamba and D. M. McDonald, “Mechanisms of adverse effects of anti-VEGF therapy for cancer,” *British Journal of Cancer*, vol. 96, no. 12, pp. 1788–1795, Jun. 2007, doi: 10.1038/sj.bjc.6603813.

[150] J. W. Ahn, D. Shalabi, L. Correa-Selm, B. Dasgeb, N. Nikbakht, and J. Cha, “Impaired wound healing secondary to bevacizumab,” *International Wound Journal*, May 2019, doi: 10.1111/iwj.13139.

[151] “UpToDate,” *UpToDate*. https://www.uptodate.com/contents/image?imageKey=HEME%2F73373&topicKey=ONC%2F14250&source=outline_link

[152] H. Izzedine, C. Massard, J.-P. Spano, F. Goldwasser, D. Khayat, and J.-C. Soria, “VEGF signalling inhibition-induced proteinuria: Mechanisms, significance and management,” *European Journal of Cancer*, vol. 46, no. 2, pp. 439–448, Jan. 2010, doi: 10.1016/j.ejca.2009.11.001.

[153] V. Eremina *et al.*, “VEGF Inhibition and Renal Thrombotic Microangiopathy,” *The New England Journal of Medicine*, vol. 358, no. 11, pp. 1129–1136, Mar. 2008, doi: 10.1056/nejmoa0707330.

[154] K. Bajbouj, R. K. Ramakrishnan, and Q. Hamid, “Role of Matrix Metalloproteinases in Angiogenesis and Its Implications in Asthma,” *Journal of Immunology Research*, vol. 2021, pp. 1–12, Feb. 2021, doi: 10.1155/2021/6645072.

[155] A. Page-McCaw, A. J. Ewald, and Z. Werb, “Matrix metalloproteinases and the regulation of tissue remodelling,” *Nature Reviews Molecular Cell Biology*, vol. 8, no. 3, pp. 221–233, Mar. 2007, doi: 10.1038/nrm2125.

[156] N. Reunanen, “Matrix Metalloproteinases in Cancer Cell Invasion,” *Madame Curie Bioscience Database - NCBI Bookshelf*, 2013. https://www.ncbi.nlm.nih.gov/books/NBK6598/

[157] E.-S. E. Mehana, A. F. Khafaga, and S. S. Elblehi, “The role of matrix metalloproteinases in osteoarthritis pathogenesis: An updated review,” *Life Sciences*, vol. 234, p. 116786, Oct. 2019, doi: 10.1016/j.lfs.2019.116786.

[158] P. Liu, M. Sun, and S. Sader, “Matrix metalloproteinases in cardiovascular disease,” *Canadian Journal of Cardiology*, vol. 22, pp. 25B-30B, Feb. 2006, doi: 10.1016/s0828-282x(06)70983-7.

[159] M. S. Hayden, A. R. West, and S. Ghosh, “NF-κB and the immune response,” *Oncogene*, vol. 25, no. 51, pp. 6758–6780, Oct. 2006, doi: 10.1038/sj.onc.1209943.

[160] T. Liu, L. Zhang, D. Joo, and S. C. Sun, “NF-κB signaling in inflammation,” *Signal Transduction and Targeted Therapy*, vol. 2, no. 1, Jul. 2017, doi: 10.1038/sigtrans.2017.23.

[161] Z. J. Xia *et al.*, “Inhibition of NF-κB signaling pathway induces apoptosis and suppresses proliferation and angiogenesis of human fibroblast-like synovial cells in rheumatoid arthritis,” *Medicine*, vol. 97, no. 23, p. e10920, Jun. 2018, doi: 10.1097/md.0000000000010920.

[162] T. Zhang, C. Ma, Z. Zhang, H. Zhang, and H. Hu, “NF-κB signaling in inflammation and cancer,” *MedComm*, vol. 2, no. 4, pp. 618–653, Dec. 2021, doi: 10.1002/mco2.104.

[163] B. D. Liebelt, G. Finocchiaro, and A. B. Heimberger, “Principles of immunotherapy,” in *Handbook of Clinical Neurology*, Elsevier BV, 2016, pp. 163–181. doi: 10.1016/b978-0-12-802997-8.00010-4.

[164] H. S. Mchaourab, J. A. Godar, and P. L. Stewart, “Structure and Mechanism of Protein Stability Sensors: Chaperone Activity of Small Heat Shock Proteins,” *Biochemistry*, vol. 48, no. 18, pp. 3828–3837, May 2009, doi: 10.1021/bi900212j.

[165] S. Chatterjee and T. F. Burns, “Targeting Heat Shock Proteins in Cancer: A Promising Therapeutic Approach,” *International Journal of Molecular Sciences*, vol. 18, no. 9, p. 1978, Sep. 2017, doi: 10.3390/ijms18091978.

[166] S. Qian, Z. Wei, W. Yang, J. Huang, Y. Yang, and J. Wang, “The role of BCL-2 family proteins in regulating apoptosis and cancer therapy,” *Frontiers in Oncology*, vol. 12, Oct. 2022, doi: 10.3389/fonc.2022.985363.

[167] M. García-Aranda, E. Pérez-Ruiz, and M. Redondo, “Bcl-2 Inhibition to Overcome Resistance to Chemo- and Immunotherapy,” *International Journal of Molecular Sciences*, vol. 19, no. 12, p. 3950, Dec. 2018, doi: 10.3390/ijms19123950.

[168] S. F. Zerp *et al.*, “Targeting anti-apoptotic Bcl-2 by AT-101 to increase radiation efficacy: data from in vitro and clinical pharmacokinetic studies in head and neck cancer,” *Radiation Oncology*, vol. 10, no. 1, Jul. 2015, doi: 10.1186/s13014-015-0474-9.